

SCHEMA IFC2X3_CV_V2_0;

(*
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Contents:

full IFC sub schema EXPRESS definitions for the IFC2X3 Coordination View Version 2.0
- ONLY released for reference and validation purposes
- SHALL NOT be used to compile and implement this sub schema
- IFC exchange files with FILE_SCHEMA(('IFC2X3_CV_V2_0')); are not valid

express longform distribution

- compiled for EXPRESS version 1 technical corrigendum 2

Issue date:

February 26, 2013

Development Team:

- Modeling Support Group of buildingSMART International
- Implementation Support Group of buildingSMART International

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- based on entity list of Coordination View Version 2.0 (v1.6 of February 26, 2013)
compiled by Karl-Heinz Haefele karl-heinz.haefele@kit.edu

Entities, not allowed to be instantiated are declared ABSTRACT:

- IfcTypeObject as Supertype of IfcTypeProduct
- IfcTypeProduct as Supertype of IfcElementType
- IfcProductRepresentation as Supertype of IfcProductDefinitionShape and IfcMaterialDefinitionRepresentation
- IfcRepresentation as Supertype of IfcStyleModel and IfcShapeModel
- IfcRepresentationContext as Supertype of IfcGeometricRepresentationContext
- IfcRelAssociates as supertype of IfcRelAssociatesClassification and IfcRelAssociatesMaterial

Changes in version 1.1 (compared to version 1.0):

- IfcBooleanResult is no longer abstract
- IfcTypeProduct is no longer abstract
- IfcVibrationIsolatorType removed

Changes in version 1.2 (compared to version 1.1):

- IfcSurfaceCurveSweptAreaSolid removed
- IfcSizeSelect added for usage in IfcCurveStyle

Changes in version 1.3 (compared to version 1.2):

- IfcRelCoversBldgElements added

*)

TYPE IfcAbsorbedDoseMeasure = REAL;
END_TYPE;

TYPE IfcAccelerationMeasure = REAL;

END_TYPE;

TYPE IfcAmountOfSubstanceMeasure = REAL;
END_TYPE;

TYPE IfcAngularVelocityMeasure = REAL;
END_TYPE;

TYPE IfcAreaMeasure = REAL;
END_TYPE;

TYPE IfcBoolean = BOOLEAN;
END_TYPE;

TYPE IfcComplexNumber = ARRAY [1:2] OF REAL;
END_TYPE;

TYPE IfcCompoundPlaneAngleMeasure = LIST [3:4] OF INTEGER;
WHERE
 WR1 : { -360 <= SELF[1] < 360 };
 WR2 : { -60 <= SELF[2] < 60 };
 WR3 : { -60 <= SELF[3] < 60 };
 WR4 : ((SELF[1] >= 0) AND (SELF[2] >= 0) AND (SELF[3] >= 0)) OR ((SELF[1] <= 0) AND
 (SELF[2] <= 0) AND (SELF[3] <= 0));
END_TYPE;

TYPE IfcContextDependentMeasure = REAL;
END_TYPE;

TYPE IfcCountMeasure = NUMBER;
END_TYPE;

TYPE IfcCurvatureMeasure = REAL;
END_TYPE;

TYPE IfcDescriptiveMeasure = STRING;
END_TYPE;

TYPE IfcDimensionCount = INTEGER;
WHERE
 WR1 : { 0 < SELF <= 3 };
END_TYPE;

TYPE IfcDoseEquivalentMeasure = REAL;
END_TYPE;

TYPE IfcDynamicViscosityMeasure = REAL;
END_TYPE;

TYPE IfcElectricCapacitanceMeasure = REAL;
END_TYPE;

TYPE IfcElectricChargeMeasure = REAL;
END_TYPE;

TYPE IfcElectricConductanceMeasure = REAL;
END_TYPE;

```
TYPE IfcElectricCurrentMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcElectricResistanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcElectricVoltageMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcEnergyMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcForceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcFrequencyMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcGloballyUniqueId = STRING(22) FIXED;  
END_TYPE;
```

```
TYPE IfcHeatFluxDensityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcHeatingValueMeasure = REAL;  
WHERE  
    WR1 : SELF > 0.;  
END_TYPE;
```

```
TYPE IfcIdentifier = STRING;  
END_TYPE;
```

```
TYPE IfcIlluminanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcInductanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcInteger = INTEGER;  
END_TYPE;
```

```
TYPE IfcIntegerCountRateMeasure = INTEGER;  
END_TYPE;
```

```
TYPE IfcIonConcentrationMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcIsothermalMoistureCapacityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcKinematicViscosityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLabel = STRING;  
END_TYPE;
```

```
TYPE IfcLengthMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLinearForceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLinearMomentMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLinearStiffnessMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLinearVelocityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLogical = LOGICAL;  
END_TYPE;
```

```
TYPE IfcLuminousFluxMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLuminousIntensityDistributionMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcLuminousIntensityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMagneticFluxDensityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMagneticFluxMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMassDensityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMassFlowRateMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMassMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMassPerLengthMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcModulusOfElasticityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcModulusOfLinearSubgradeReactionMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcModulusOfRotationalSubgradeReactionMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcModulusOfSubgradeReactionMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMoistureDiffusivityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMolecularWeightMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMomentOfInertiaMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcMonetaryMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcNormalisedRatioMeasure = IfcRatioMeasure;  
  WHERE  
    WR1 : {0.0 <= SELF <= 1.0};  
END_TYPE;
```

```
TYPE IfcNumericMeasure = NUMBER;  
END_TYPE;
```

```
TYPE IfcPHMeasure = REAL;  
  WHERE  
    WR21 : {0.0 <= SELF <= 14.0};  
END_TYPE;
```

```
TYPE IfcParameterValue = REAL;  
END_TYPE;
```

```
TYPE IfcPlanarForceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcPlaneAngleMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcPositiveLengthMeasure = IfcLengthMeasure;  
  WHERE  
    WR1 : SELF > 0.;  
END_TYPE;
```

```
TYPE IfcPositivePlaneAngleMeasure = IfcPlaneAngleMeasure;  
  WHERE  
    WR1 : SELF > 0.;  
END_TYPE;
```

```
TYPE IfcPositiveRatioMeasure = IfcRatioMeasure;  
  WHERE  
    WR1 : SELF > 0.;  
END_TYPE;
```

```
TYPE IfcPowerMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcPressureMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcRadioActivityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcRatioMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcReal = REAL;  
END_TYPE;
```

```
TYPE IfcRotationalFrequencyMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcRotationalMassMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcRotationalStiffnessMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSectionModulusMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSectionalAreaIntegralMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcShearModulusMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSolidAngleMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSoundPowerMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSoundPressureMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSpecificHeatCapacityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcSpecularExponent = REAL;  
END_TYPE;
```

```
TYPE IfcSpecularRoughness = REAL;  
WHERE  
    WR1 : {0.0 <= SELF <= 1.0};  
END_TYPE;
```

```
TYPE IfcStrippedOptional = BOOLEAN;  
END_TYPE;
```

```
TYPE IfcTemperatureGradientMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcText = STRING;  
END_TYPE;
```

```
TYPE IfcThermalAdmittanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcThermalConductivityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcThermalExpansionCoefficientMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcThermalResistanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcThermalTransmittanceMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcThermodynamicTemperatureMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcTimeMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcTimeStamp = INTEGER;  
END_TYPE;
```

```
TYPE IfcTorqueMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcVaporPermeabilityMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcVolumeMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcVolumetricFlowRateMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcWarpingConstantMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcWarpingMomentMeasure = REAL;  
END_TYPE;
```

```
TYPE IfcActuatorTypeEnum = ENUMERATION OF  
    (ELECTRICACTIONATOR  
    ,HANDOPERATEDACTUATOR  
    ,HYDRAULICACTIONATOR  
    ,PNEUMATICACTIONATOR  
    ,THERMOSTATICACTIONATOR  
    ,USERDEFINED  
    ,NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcAddressTypeEnum = ENUMERATION OF  
    (OFFICE  
    ,SITE  
    ,HOME  
    ,DISTRIBUTIONPOINT  
    ,USERDEFINED);  
END_TYPE;
```

```

TYPE IfcAirTerminalBoxTypeEnum = ENUMERATION OF
  (CONSTANTFLOW
  ,VARIABLEFLOWPRESSUREDEPENDANT
  ,VARIABLEFLOWPRESSUREINDEPENDANT
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;

```

```

TYPE IfcAirTerminalTypeEnum = ENUMERATION OF
  (GRILLE
  ,REGISTER
  ,DIFFUSER
  ,EYEBALL
  ,IRIS
  ,LINEARGRILLE
  ,LINEARDIFFUSER
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;

```

```

TYPE IfcAirToAirHeatRecoveryTypeEnum = ENUMERATION OF
  (FIXEDPLATECOUNTERFLOWEXCHANGER
  ,FIXEDPLATECROSSFLOWEXCHANGER
  ,FIXEDPLATEPARALLELFLOWEXCHANGER
  ,ROTARYWHEEL
  ,RUNAROUNDCOILLOOP
  ,HEATPIPE
  ,TWINTOWERENTHALPYRECOVERYLOOPS
  ,THERMOSIPHONSEALEDTUBEHEATEXCHANGERS
  ,THERMOSIPHONCOILTYPEHEATEXCHANGERS
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;

```

```

TYPE IfcAlarmTypeEnum = ENUMERATION OF
  (BELL
  ,BREAKGLASSBUTTON
  ,LIGHT
  ,MANUALPULLBOX
  ,SIREN
  ,WHISTLE
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;

```

```

TYPE IfcAssemblyPlaceEnum = ENUMERATION OF
  (SITE
  ,FACTORY
  ,NOTDEFINED);
END_TYPE;

```

```

TYPE IfcBeamTypeEnum = ENUMERATION OF
  (BEAM
  ,JOIST
  ,LINTEL
  ,T_BEAM
  ,USERDEFINED

```

```
, NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcBoilerTypeEnum = ENUMERATION OF  
    (WATER  
    , STEAM  
    , USERDEFINED  
    , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcBooleanOperator = ENUMERATION OF  
    (UNION  
    , INTERSECTION  
    , DIFFERENCE);  
END_TYPE;
```

```
TYPE IfcBuildingElementProxyTypeEnum = ENUMERATION OF  
    (USERDEFINED  
    , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCableCarrierFittingTypeEnum = ENUMERATION OF  
    (BEND  
    , CROSS  
    , REDUCER  
    , TEE  
    , USERDEFINED  
    , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCableCarrierSegmentTypeEnum = ENUMERATION OF  
    (CABLELADDERSEGMENT  
    , CABLETRAYSEGMENT  
    , CABLETRUNKINGSEGMENT  
    , CONDUITSEGMENT  
    , USERDEFINED  
    , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCableSegmentTypeEnum = ENUMERATION OF  
    (CABLESEGMENT  
    , CONDUCTORSEGMENT  
    , USERDEFINED  
    , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcChangeActionEnum = ENUMERATION OF  
    (NOCHANGE  
    , MODIFIED  
    , ADDED  
    , DELETED  
    , MODIFIEDADDED  
    , MODIFIEDDELETED);  
END_TYPE;
```

```
TYPE IfcChillerTypeEnum = ENUMERATION OF  
    (AIRCOOLED
```

```
, WATERCOOLED
, HEATRECOVERY
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcCoilTypeEnum = ENUMERATION OF
```

```
(DXCOOLINGCOIL
, WATERCOOLINGCOIL
, STEAMHEATINGCOIL
, WATERHEATINGCOIL
, ELECTRICHEATINGCOIL
, GASHEATINGCOIL
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcColumnTypeEnum = ENUMERATION OF
```

```
(COLUMN
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcCompressorTypeEnum = ENUMERATION OF
```

```
(DYNAMIC
, RECIPROCATING
, ROTARY
, SCROLL
, TROCHOIDAL
, SINGLESTAGE
, BOOSTER
, OPENTYPE
, HERMETIC
, SEMIHERMETIC
, WELDEDSHELLHERMETIC
, ROLLINGPISTON
, ROTARYVANE
, SINGLESCREW
, TWINSCREW
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcCondenserTypeEnum = ENUMERATION OF
```

```
(WATERCOOLEDSHELLTUBE
, WATERCOOLEDSHELLCOIL
, WATERCOOLEDPIPEINTUBE
, WATERCOOLEDBRAZEDPLATE
, AIRCOOLED
, EVAPORATIVECOOLED
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcConnectionTypeEnum = ENUMERATION OF
```

```
(ATPATH
, ATSTART
```

```
, ATEND  
, NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcControllerTypeEnum = ENUMERATION OF  
  (FLOATING  
  , PROPORTIONAL  
  , PROPORTIONALINTEGRAL  
  , PROPORTIONALINTEGRALDERIVATIVE  
  , TIMEDTWOPOSITION  
  , TWOPOSITION  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCooledBeamTypeEnum = ENUMERATION OF  
  (ACTIVE  
  , PASSIVE  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCoolingTowerTypeEnum = ENUMERATION OF  
  (NATURALDRAFT  
  , MECHANICALINDUCEDDRAFT  
  , MECHANICALFORCEDDRAFT  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCoveringTypeEnum = ENUMERATION OF  
  (CEILING  
  , FLOORING  
  , CLADDING  
  , ROOFING  
  , INSULATION  
  , MEMBRANE  
  , SLEEVING  
  , WRAPPING  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcCurtainWallTypeEnum = ENUMERATION OF  
  (USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcDamperTypeEnum = ENUMERATION OF  
  (CONTROLDAMPER  
  , FIREDAMPER  
  , SMOKEDAMPER  
  , FIRESMOKEDAMPER  
  , BACKDRAFTDAMPER  
  , RELIEFDAMPER  
  , BLASTDAMPER  
  , GRAVITYDAMPER
```

```
, GRAVITYRELIEFDAMPER
, BALANCINGDAMPER
, FUMEHOODEXHAUST
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcDerivedUnitEnum = ENUMERATION OF
```

```
( ANGULARVELOCITYUNIT
, COMPOUNDPLANEANGLEUNIT
, DYNAMICVISCOSITYUNIT
, HEATFLUXDENSITYUNIT
, INTEGERCOUNTRATEUNIT
, ISOTHERMALMOISTURECAPACITYUNIT
, KINEMATICVISCOSITYUNIT
, LINEARVELOCITYUNIT
, MASSDENSITYUNIT
, MASSFLOWRATEUNIT
, MOISTUREDIFUSIVITYUNIT
, MOLECULARWEIGHTUNIT
, SPECIFICHEATCAPACITYUNIT
, THERMALADMITTANCEUNIT
, THERMALCONDUCTANCEUNIT
, THERMALRESISTANCEUNIT
, THERMALTRANSMITTANCEUNIT
, VAPORPERMEABILITYUNIT
, VOLUMETRICFLOWRATEUNIT
, ROTATIONALFREQUENCYUNIT
, TORQUEUNIT
, MOMENTOFINERTIAUNIT
, LINEARMOMENTUNIT
, LINEARFORCEUNIT
, PLANARFORCEUNIT
, MODULUSOFELASTICITYUNIT
, SHEARMODULUSUNIT
, LINEARSTIFFNESSUNIT
, ROTATIONALSTIFFNESSUNIT
, MODULUSOF SUBGRADEREACTIONUNIT
, ACCELERATIONUNIT
, CURVATUREUNIT
, HEATINGVALUEUNIT
, IONCONCENTRATIONUNIT
, LUMINOUSINTENSITYDISTRIBUTIONUNIT
, MASSPERLENGTHUNIT
, MODULUSOF LINEAR SUBGRADEREACTIONUNIT
, MODULUSOF ROTATIONAL SUBGRADEREACTIONUNIT
, PHUNIT
, ROTATIONALMASSUNIT
, SECTIONAREAINTEGRALUNIT
, SECTIONMODULUSUNIT
, SOUNDPOWERUNIT
, SOUNDPRESSUREUNIT
, TEMPERATUREGRADIENTUNIT
, THERMALEXPANSIONCOEFFICIENTUNIT
, WARPINGCONSTANTUNIT
, WARPINGMOMENTUNIT
, USERDEFINED);
```

END_TYPE;

```
TYPE IfcDirectionSenseEnum = ENUMERATION OF
  (POSITIVE
  ,NEGATIVE);
END_TYPE;
```

```
TYPE IfcDistributionChamberElementTypeEnum = ENUMERATION OF
  (FORMEDDUCT
  ,INSPECTIONCHAMBER
  ,INSPECTIONPIT
  ,MANHOLE
  ,METERCHAMBER
  ,SUMP
  ,TRENCH
  ,VALVECHAMBER
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;
```

```
TYPE IfcDoorPanelOperationEnum = ENUMERATION OF
  (SWINGING
  ,DOUBLE_ACTING
  ,SLIDING
  ,FOLDING
  ,REVOLVING
  ,ROLLINGUP
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;
```

```
TYPE IfcDoorPanelPositionEnum = ENUMERATION OF
  (LEFT
  ,MIDDLE
  ,RIGHT
  ,NOTDEFINED);
END_TYPE;
```

```
TYPE IfcDoorStyleConstructionEnum = ENUMERATION OF
  (ALUMINIUM
  ,HIGH_GRADE_STEEL
  ,STEEL
  ,WOOD
  ,ALUMINIUM_WOOD
  ,ALUMINIUM_PLASTIC
  ,PLASTIC
  ,USERDEFINED
  ,NOTDEFINED);
END_TYPE;
```

```
TYPE IfcDoorStyleOperationEnum = ENUMERATION OF
  (SINGLE_SWING_LEFT
  ,SINGLE_SWING_RIGHT
  ,DOUBLE_DOOR_SINGLE_SWING
  ,DOUBLE_DOOR_SINGLE_SWING_OPPOSITE_LEFT
  ,DOUBLE_DOOR_SINGLE_SWING_OPPOSITE_RIGHT
  ,DOUBLE_SWING_LEFT
```

```
, DOUBLE_SWING_RIGHT
, DOUBLE_DOOR_DOUBLE_SWING
, SLIDING_TO_LEFT
, SLIDING_TO_RIGHT
, DOUBLE_DOOR_SLIDING
, FOLDING_TO_LEFT
, FOLDING_TO_RIGHT
, DOUBLE_DOOR_FOLDING
, REVOLVING
, ROLLINGUP
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcDuctFittingTypeEnum = ENUMERATION OF
```

```
(BEND
, CONNECTOR
, ENTRY
, EXIT
, JUNCTION
, OBSTRUCTION
, TRANSITION
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcDuctSegmentTypeEnum = ENUMERATION OF
```

```
(RIGIDSEGMENT
, FLEXIBLESEGMENT
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcDuctSilencerTypeEnum = ENUMERATION OF
```

```
(FLATOVAL
, RECTANGULAR
, ROUND
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElectricApplianceTypeEnum = ENUMERATION OF
```

```
(COMPUTER
, DIRECTWATERHEATER
, DISHWASHER
, ELECTRICCOOKER
, ELECTRICHEATER
, FACSIMILE
, FREESTANDINGFAN
, FREEZER
, FRIDGE_FREEZER
, HANDDRYER
, INDIRECTWATERHEATER
, MICROWAVE
, PHOTOCOPIER
, PRINTER
, REFRIGERATOR
```

```
, RADIANTHEATER  
, SCANNER  
, TELEPHONE  
, TUMBLEDRYER  
, TV  
, VENDINGMACHINE  
, WASHINGMACHINE  
, WATERHEATER  
, WATERCOOLER  
, USERDEFINED  
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElectricFlowStorageDeviceTypeEnum = ENUMERATION OF
```

```
( BATTERY  
, CAPACITORBANK  
, HARMONICFILTER  
, INDUCTORBANK  
, UPS  
, USERDEFINED  
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElectricGeneratorTypeEnum = ENUMERATION OF
```

```
( USERDEFINED  
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElectricMotorTypeEnum = ENUMERATION OF
```

```
( DC  
, INDUCTION  
, POLYPHASE  
, RELUCTANCESYNCHRONOUS  
, SYNCHRONOUS  
, USERDEFINED  
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElectricTimeControlTypeEnum = ENUMERATION OF
```

```
( TIMECLOCK  
, TIMEDELAY  
, RELAY  
, USERDEFINED  
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcElementAssemblyTypeEnum = ENUMERATION OF
```

```
( ACCESSORY_ASSEMBLY  
, ARCH  
, BEAM_GRID  
, BRACED_FRAME  
, GIRDER  
, REINFORCEMENT_UNIT  
, RIGID_FRAME  
, SLAB_FIELD  
, TRUSS  
, USERDEFINED
```

```
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcElementCompositionEnum = ENUMERATION OF
  (COMPLEX
  , ELEMENT
  , PARTIAL);
END_TYPE;
```

```
TYPE IfcEvaporativeCoolerTypeEnum = ENUMERATION OF
  (DIRECTEVAPORATIVERANDOMMEDIAAIRCOOLER
  , DIRECTEVAPORATIVERIGIDMEDIAAIRCOOLER
  , DIRECTEVAPORATIVESLINGERSPACKAGEDAIRCOOLER
  , DIRECTEVAPORATIVEPACKAGEDROTARYAIRCOOLER
  , DIRECTEVAPORATIVEAIRWASHER
  , INDIRECTEVAPORATIVEPACKAGEAIRCOOLER
  , INDIRECTEVAPORATIVEWETCOIL
  , INDIRECTEVAPORATIVECOOLINGTOWERORCOILCOOLER
  , INDIRECTDIRECTCOMBINATION
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcEvaporatorTypeEnum = ENUMERATION OF
  (DIRECTEXPANSIONSHELLANDTUBE
  , DIRECTEXPANSIONTUBEINTUBE
  , DIRECTEXPANSIONBRAZEDPLATE
  , FLOODEDSHELLANDTUBE
  , SHELLANDCOIL
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFanTypeEnum = ENUMERATION OF
  (CENTRIFUGALFORWARDCURVED
  , CENTRIFUGALRADIAL
  , CENTRIFUGALBACKWARDINCLINEDCURVED
  , CENTRIFUGALAIRFOIL
  , TUBEAXIAL
  , VANEAXIAL
  , PROPELLORAXIAL
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFilterTypeEnum = ENUMERATION OF
  (AIRPARTICLEFILTER
  , ODORFILTER
  , OILFILTER
  , STRAINER
  , WATERFILTER
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFireSuppressionTerminalTypeEnum = ENUMERATION OF
  (BREECHINGINLET
```

```
, FIREHYDRANT
, HOSEREEL
, SPRINKLER
, SPRINKLERDEFLECTOR
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFlowDirectionEnum = ENUMERATION OF
(SOURCE
, SINK
, SOURCEANDSINK
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFlowInstrumentTypeEnum = ENUMERATION OF
(PRESSUREGAUGE
, THERMOMETER
, AMMETER
, FREQUENCYMETER
, POWERFACTORMETER
, PHASEANGLEMETER
, VOLTMETER_PEAK
, VOLTMETER_RMS
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFlowMeterTypeEnum = ENUMERATION OF
(ELECTRICMETER
, ENERGYMETER
, FLOWMETER
, GASMETER
, OILMETER
, WATERMETER
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcFootingTypeEnum = ENUMERATION OF
(FOOTING_BEAM
, PAD_FOOTING
, PILE_CAP
, STRIP_FOOTING
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcGeometricProjectionEnum = ENUMERATION OF
(GRAPH_VIEW
, SKETCH_VIEW
, MODEL_VIEW
, PLAN_VIEW
, REFLECTED_PLAN_VIEW
, SECTION_VIEW
, ELEVATION_VIEW
, USERDEFINED
```

```
, NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcHeatExchangerTypeEnum = ENUMERATION OF  
  ( PLATE  
  , SHELLANDTUBE  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcHumidifierTypeEnum = ENUMERATION OF  
  ( STEAMINJECTION  
  , ADIABATICAIRWASHER  
  , ADIABATICPAN  
  , ADIABATICWETTEDELEMENT  
  , ADIABATICATOMIZING  
  , ADIABATICULTRASONIC  
  , ADIABATICRIGIDMEDIA  
  , ADIABATICCOMPRESSEDFAIRNOZZLE  
  , ASSISTEDELECTRIC  
  , ASSISTEDNATURALGAS  
  , ASSISTEDPROPANE  
  , ASSISTEDBUTANE  
  , ASSISTEDSTEAM  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcInternalOrExternalEnum = ENUMERATION OF  
  ( INTERNAL  
  , EXTERNAL  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcJunctionBoxTypeEnum = ENUMERATION OF  
  ( USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcLampTypeEnum = ENUMERATION OF  
  ( COMPACTFLUORESCENT  
  , FLUORESCENT  
  , HIGHPRESSUREMERCURY  
  , HIGHPRESSURESODIUM  
  , METALHALIDE  
  , TUNGSTENFILAMENT  
  , USERDEFINED  
  , NOTDEFINED);  
END_TYPE;
```

```
TYPE IfcLayerSetDirectionEnum = ENUMERATION OF  
  ( AXIS1  
  , AXIS2  
  , AXIS3);  
END_TYPE;
```

```
TYPE IfcLightFixtureTypeEnum = ENUMERATION OF
```

```

    (POINTSOURCE
    , DIRECTIONSOURCE
    , USERDEFINED
    , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcMemberTypeEnum = ENUMERATION OF
    (BRACE
    , CHORD
    , COLLAR
    , MEMBER
    , MULLION
    , PLATE
    , POST
    , PURLIN
    , RAFTER
    , STRINGER
    , STRUT
    , STUD
    , USERDEFINED
    , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcMotorConnectionTypeEnum = ENUMERATION OF
    (BELTDRIVE
    , COUPLING
    , DIRECTDRIVE
    , USERDEFINED
    , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcNullStyle = ENUMERATION OF
    (NULL);
END_TYPE;

```

```

TYPE IfcObjectTypeEnum = ENUMERATION OF
    (PRODUCT
    , PROCESS
    , CONTROL
    , RESOURCE
    , ACTOR
    , GROUP
    , PROJECT
    , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcOutletTypeEnum = ENUMERATION OF
    (AUDIOVISUALOUTLET
    , COMMUNICATIONSOUTLET
    , POWEROUTLET
    , USERDEFINED
    , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcPileConstructionEnum = ENUMERATION OF
    (CAST_IN_PLACE
    , COMPOSITE

```

```
, PRECAST_CONCRETE
, PREFAB_STEEL
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcPileTypeEnum = ENUMERATION OF
( COHESION
, FRICTION
, SUPPORT
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcPipeFittingTypeEnum = ENUMERATION OF
( BEND
, CONNECTOR
, ENTRY
, EXIT
, JUNCTION
, OBSTRUCTION
, TRANSITION
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcPipeSegmentTypeEnum = ENUMERATION OF
( FLEXIBLESEGMENT
, RIGIDSEGMENT
, GUTTER
, SPOOL
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcPlateTypeEnum = ENUMERATION OF
( CURTAIN_PANEL
, SHEET
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcProfileTypeEnum = ENUMERATION OF
( CURVE
, AREA);
END_TYPE;
```

```
TYPE IfcProtectiveDeviceTypeEnum = ENUMERATION OF
( FUSEDISCONNECTOR
, CIRCUITBREAKER
, EARTHFAILUREDEVICE
, RESIDUALCURRENTCIRCUITBREAKER
, RESIDUALCURRENTSWITCH
, VARISTOR
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcPumpTypeEnum = ENUMERATION OF
  (CIRCULATOR
  , ENDSUCTION
  , SPLITCASE
  , VERTICALINLINE
  , VERTICALTURBINE
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcRailingTypeEnum = ENUMERATION OF
  (HANDRAIL
  , GUARDRAIL
  , BALUSTRADE
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcRampFlightTypeEnum = ENUMERATION OF
  (STRAIGHT
  , SPIRAL
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcRampTypeEnum = ENUMERATION OF
  (STRAIGHT_RUN_RAMP
  , TWO_STRAIGHT_RUN_RAMP
  , QUARTER_TURN_RAMP
  , TWO_QUARTER_TURN_RAMP
  , HALF_TURN_RAMP
  , SPIRAL_RAMP
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcReflectanceMethodEnum = ENUMERATION OF
  (BLINN
  , FLAT
  , GLASS
  , MATT
  , METAL
  , MIRROR
  , PHONG
  , PLASTIC
  , STRAUSS
  , NOTDEFINED);
END_TYPE;
```

```
TYPE IfcReinforcingBarRoleEnum = ENUMERATION OF
  (MAIN
  , SHEAR
  , LIGATURE
  , STUD
  , PUNCHING
  , EDGE
```

```
, RING
, USERDEFINED
, NOTDEFINED);
END_TYPE;

TYPE IfcReinforcingBarSurfaceEnum = ENUMERATION OF
(PLAIN
, TEXTURED);
END_TYPE;
```

```
TYPE IfcRoleEnum = ENUMERATION OF
(SUPPLIER
, MANUFACTURER
, CONTRACTOR
, SUBCONTRACTOR
, ARCHITECT
, STRUCTURALENGINEER
, COSTENGINEER
, CLIENT
, BUILDINGOWNER
, BUILDINGOPERATOR
, MECHANICALENGINEER
, ELECTRICALENGINEER
, PROJECTMANAGER
, FACILITIESMANAGER
, CIVILENGINEER
, COMMISSIONINGENGINEER
, ENGINEER
, OWNER
, CONSULTANT
, CONSTRUCTIONMANAGER
, FIELDCONSTRUCTIONMANAGER
, RESELLER
, USERDEFINED);
END_TYPE;
```

```
TYPE IfcRoofTypeEnum = ENUMERATION OF
(FLAT_ROOF
, SHED_ROOF
, GABLE_ROOF
, HIP_ROOF
, HIPPED_GABLE_ROOF
, GAMBREL_ROOF
, MANSARD_ROOF
, BARREL_ROOF
, RAINBOW_ROOF
, BUTTERFLY_ROOF
, PAVILION_ROOF
, DOME_ROOF
, FREEFORM
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcSIPrefix = ENUMERATION OF
(EXA
, PETA
, TERA
```

```
, GIGA  
, MEGA  
, KILO  
, HECTO  
, DECA  
, DECI  
, CENTI  
, MILLI  
, MICRO  
, NANO  
, PICO  
, FEMTO  
, ATTO);  
END_TYPE;
```

```
TYPE IfcSIUnitName = ENUMERATION OF
```

```
( AMPERE  
, BECQUEREL  
, CANDELA  
, COULOMB  
, CUBIC_METRE  
, DEGREE_CELSIUS  
, FARAD  
, GRAM  
, GRAY  
, HENRY  
, HERTZ  
, JOULE  
, KELVIN  
, LUMEN  
, LUX  
, METRE  
, MOLE  
, NEWTON  
, OHM  
, PASCAL  
, RADIAN  
, SECOND  
, SIEMENS  
, SIEVERT  
, SQUARE_METRE  
, STERADIAN  
, TESLA  
, VOLT  
, WATT  
, WEBER);  
END_TYPE;
```

```
TYPE IfcSanitaryTerminalTypeEnum = ENUMERATION OF
```

```
( BATH  
, BIDET  
, CISTERN  
, SHOWER  
, SINK  
, SANITARYFOUNTAIN  
, TOILETPAN  
, URINAL
```

```
, WASHHANDBASIN
, WCSEAT
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcSensorTypeEnum = ENUMERATION OF
( CO2SENSOR
, FIRESSENSOR
, FLOWSSENSOR
, GASSENSOR
, HEATSSENSOR
, HUMIDITYSENSOR
, LIGHTSENSOR
, MOISTURESENSOR
, MOVEMENTSENSOR
, PRESSURESENSOR
, SMOKESENSOR
, SOUNDSSENSOR
, TEMPERATURESENSOR
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcSlabTypeEnum = ENUMERATION OF
( FLOOR
, ROOF
, LANDING
, BASESLAB
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcSpaceHeaterTypeEnum = ENUMERATION OF
( SECTIONALRADIATOR
, PANELRADIATOR
, TUBULARRADIATOR
, CONVECTOR
, BASEBOARDHEATER
, FINNEDTUBEUNIT
, UNITHEATER
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcSpaceTypeEnum = ENUMERATION OF
( USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcStackTerminalTypeEnum = ENUMERATION OF
( BIRDCAGE
, COWL
, RAINWATERHOPPER
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```

TYPE IfcStairFlightTypeEnum = ENUMERATION OF
  (STRAIGHT
  , WINDER
  , SPIRAL
  , CURVED
  , FREEFORM
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcStairTypeEnum = ENUMERATION OF
  (STRAIGHT_RUN_STAIR
  , TWO_STRAIGHT_RUN_STAIR
  , QUARTER_WINDING_STAIR
  , QUARTER_TURN_STAIR
  , HALF_WINDING_STAIR
  , HALF_TURN_STAIR
  , TWO_QUARTER_WINDING_STAIR
  , TWO_QUARTER_TURN_STAIR
  , THREE_QUARTER_WINDING_STAIR
  , THREE_QUARTER_TURN_STAIR
  , SPIRAL_STAIR
  , DOUBLE_RETURN_STAIR
  , CURVED_RUN_STAIR
  , TWO_CURVED_RUN_STAIR
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcStateEnum = ENUMERATION OF
  (READWRITE
  , READONLY
  , LOCKED
  , READWRITELOCKED
  , READONLYLOCKED);
END_TYPE;

```

```

TYPE IfcSurfaceSide = ENUMERATION OF
  (POSITIVE
  , NEGATIVE
  , BOTH);
END_TYPE;

```

```

TYPE IfcSwitchingDeviceTypeEnum = ENUMERATION OF
  (CONTACTOR
  , EMERGENCYSTOP
  , STARTER
  , SWITCHDISCONNECTOR
  , TOGGLESWITCH
  , USERDEFINED
  , NOTDEFINED);
END_TYPE;

```

```

TYPE IfcTankTypeEnum = ENUMERATION OF
  (PREFORMED
  , SECTIONAL

```

```
, EXPANSION
, PRESSUREVESSEL
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcTendonTypeEnum = ENUMERATION OF
(STRAND
, WIRE
, BAR
, COATED
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcTransformerTypeEnum = ENUMERATION OF
(CURRENT
, FREQUENCY
, VOLTAGE
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcTransitionCode = ENUMERATION OF
(DISCONTINUOUS
, CONTINUOUS
, CONTSAMEGRADIENT
, CONTSAMEGRADIENTSAMECURVATURE);
END_TYPE;
```

```
TYPE IfcTransportElementTypeEnum = ENUMERATION OF
(ELEVATOR
, ESCALATOR
, MOVINGWALKWAY
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcTrimmingPreference = ENUMERATION OF
(CARTESIAN
, PARAMETER
, UNSPECIFIED);
END_TYPE;
```

```
TYPE IfcTubeBundleTypeEnum = ENUMERATION OF
(FINNED
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcUnitEnum = ENUMERATION OF
(ABSORBEDDOSEUNIT
, AMOUNTOFSUBSTANCEUNIT
, AREAUNIT
, DOSEEQUIVALENTUNIT
, ELECTRICCAPACITANCEUNIT
, ELECTRICCHARGEUNIT
```

```
, ELECTRICCONDUCTANCEUNIT
, ELECTRICCURRENTUNIT
, ELECTRICRESISTANCEUNIT
, ELECTRICVOLTAGEUNIT
, ENERGYUNIT
, FORCEUNIT
, FREQUENCYUNIT
, ILLUMINANCEUNIT
, INDUCTANCEUNIT
, LENGTHUNIT
, LUMINOUSFLUXUNIT
, LUMINOUSINTENSITYUNIT
, MAGNETICFLUXDENSITYUNIT
, MAGNETICFLUXUNIT
, MASSUNIT
, PLANEANGLEUNIT
, POWERUNIT
, PRESSUREUNIT
, RADIOACTIVITYUNIT
, SOLIDANGLEUNIT
, THERMODYNAMICTEMPERATUREUNIT
, TIMEUNIT
, VOLUMEUNIT
, USERDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcUnitaryEquipmentTypeEnum = ENUMERATION OF
```

```
(AIRHANDLER
, AIRCONDITIONINGUNIT
, SPLITSYSTEM
, ROOFTOPUNIT
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcValveTypeEnum = ENUMERATION OF
```

```
(AIRRELEASE
, ANTIVACUUM
, CHANGEOVER
, CHECK
, COMMISSIONING
, DIVERTING
, DRAWOFFCOCK
, DOUBLECHECK
, DOUBLEREGULATING
, FAUCET
, FLUSHING
, GASCOCK
, GASTAP
, ISOLATING
, MIXING
, PRESSUREREDUCING
, PRESSURERELIEF
, REGULATING
, SAFETYCUTOFF
, STEAMTRAP
, STOPCOCK
```

```
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcWallTypeEnum = ENUMERATION OF
( STANDARD
, POLYGONAL
, SHEAR
, ELEMENTEDWALL
, PLUMBINGWALL
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcWasteTerminalTypeEnum = ENUMERATION OF
( FLOORTRAP
, FLOORWASTE
, GULLYSUMP
, GULLYTRAP
, GREASEINTERCEPTOR
, OILINTERCEPTOR
, PETROLINTERCEPTOR
, ROOFDRAIN
, WASTEDISPOSALUNIT
, WASTETRAP
, USERDEFINED
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcWindowPanelOperationEnum = ENUMERATION OF
( SIDEHUNGRIGHTHAND
, SIDEHUNGLEFTHAND
, TILTANDTURNRIGHTHAND
, TILTANDTURNLEFTHAND
, TOPHUNG
, BOTTOMHUNG
, PIVOTHORIZONTAL
, PIVOTVERTICAL
, SLIDINGHORIZONTAL
, SLIDINGVERTICAL
, REMOVABLECASEMENT
, FIXEDCASEMENT
, OTHEROPERATION
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcWindowPanelPositionEnum = ENUMERATION OF
( LEFT
, MIDDLE
, RIGHT
, BOTTOM
, TOP
, NOTDEFINED);
END_TYPE;
```

```
TYPE IfcWindowStyleConstructionEnum = ENUMERATION OF
( ALUMINIUM
```

```
, HIGH_GRADE_STEEL
, STEEL
, WOOD
, ALUMINIUM_WOOD
, PLASTIC
, OTHER_CONSTRUCTION
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcWindowStyleOperationEnum = ENUMERATION OF
```

```
( SINGLE_PANEL
, DOUBLE_PANEL_VERTICAL
, DOUBLE_PANEL_HORIZONTAL
, TRIPLE_PANEL_VERTICAL
, TRIPLE_PANEL_BOTTOM
, TRIPLE_PANEL_TOP
, TRIPLE_PANEL_LEFT
, TRIPLE_PANEL_RIGHT
, TRIPLE_PANEL_HORIZONTAL
, USERDEFINED
, NOTDEFINED);
```

```
END_TYPE;
```

```
TYPE IfcAxis2Placement = SELECT
```

```
( IfcAxis2Placement2D
, IfcAxis2Placement3D);
```

```
END_TYPE;
```

```
TYPE IfcBooleanOperand = SELECT
```

```
( IfcBooleanResult
, IfcHalfSpaceSolid
, IfcSolidModel);
```

```
END_TYPE;
```

```
TYPE IfcClassificationNotationSelect = SELECT
```

```
( IfcClassificationReference);
```

```
END_TYPE;
```

```
TYPE IfcColour = SELECT
```

```
( IfcColourSpecification);
```

```
END_TYPE;
```

```
TYPE IfcColourOrFactor = SELECT
```

```
( IfcColourRgb
, IfcNormalisedRatioMeasure);
```

```
END_TYPE;
```

```
TYPE IfcCurveOrEdgeCurve = SELECT
```

```
( IfcBoundedCurve);
```

```
END_TYPE;
```

```
TYPE IfcDerivedMeasureValue = SELECT
```

```
( IfcAbsorbedDoseMeasure
, IfcAccelerationMeasure
, IfcAngularVelocityMeasure
, IfcCompoundPlaneAngleMeasure
, IfcCurvatureMeasure
```

, IfcDoseEquivalentMeasure
, IfcDynamicViscosityMeasure
, IfcElectricCapacitanceMeasure
, IfcElectricChargeMeasure
, IfcElectricConductanceMeasure
, IfcElectricResistanceMeasure
, IfcElectricVoltageMeasure
, IfcEnergyMeasure
, IfcForceMeasure
, IfcFrequencyMeasure
, IfcHeatFluxDensityMeasure
, IfcHeatingValueMeasure
, IfcIlluminanceMeasure
, IfcInductanceMeasure
, IfcIntegerCountRateMeasure
, IfcIonConcentrationMeasure
, IfcIsothermalMoistureCapacityMeasure
, IfcKinematicViscosityMeasure
, IfcLinearForceMeasure
, IfcLinearMomentMeasure
, IfcLinearStiffnessMeasure
, IfcLinearVelocityMeasure
, IfcLuminousFluxMeasure
, IfcLuminousIntensityDistributionMeasure
, IfcMagneticFluxDensityMeasure
, IfcMagneticFluxMeasure
, IfcMassDensityMeasure
, IfcMassFlowRateMeasure
, IfcMassPerLengthMeasure
, IfcModulusOfElasticityMeasure
, IfcModulusOfLinearSubgradeReactionMeasure
, IfcModulusOfRotationalSubgradeReactionMeasure
, IfcModulusOfSubgradeReactionMeasure
, IfcMoistureDiffusivityMeasure
, IfcMolecularWeightMeasure
, IfcMomentOfInertiaMeasure
, IfcMonetaryMeasure
, IfcPHMeasure
, IfcPlanarForceMeasure
, IfcPowerMeasure
, IfcPressureMeasure
, IfcRadioActivityMeasure
, IfcRotationalFrequencyMeasure
, IfcRotationalMassMeasure
, IfcRotationalStiffnessMeasure
, IfcSectionModulusMeasure
, IfcSectionalAreaIntegralMeasure
, IfcShearModulusMeasure
, IfcSoundPowerMeasure
, IfcSoundPressureMeasure
, IfcSpecificHeatCapacityMeasure
, IfcTemperatureGradientMeasure
, IfcThermalAdmittanceMeasure
, IfcThermalConductivityMeasure
, IfcThermalExpansionCoefficientMeasure
, IfcThermalResistanceMeasure
, IfcThermalTransmittanceMeasure

```
, IfcTimeStamp
, IfcTorqueMeasure
, IfcVaporPermeabilityMeasure
, IfcVolumetricFlowRateMeasure
, IfcWarpingConstantMeasure
, IfcWarpingMomentMeasure);
END_TYPE;
```

```
TYPE IfcGeometricSetSelect = SELECT
(IfcCurve
, IfcPoint
, IfcSurface);
END_TYPE;
```

```
TYPE IfcLayeredItem = SELECT
(IfcRepresentation
, IfcRepresentationItem);
END_TYPE;
```

```
TYPE IfcMaterialSelect = SELECT
(IfcMaterial
, IfcMaterialLayer
, IfcMaterialLayerSet
, IfcMaterialLayerSetUsage
, IfcMaterialList);
END_TYPE;
```

```
TYPE IfcMeasureValue = SELECT
(IfcAmountOfSubstanceMeasure
, IfcAreaMeasure
, IfcComplexNumber
, IfcContextDependentMeasure
, IfcCountMeasure
, IfcDescriptiveMeasure
, IfcElectricCurrentMeasure
, IfcLengthMeasure
, IfcLuminousIntensityMeasure
, IfcMassMeasure
, IfcNormalisedRatioMeasure
, IfcNumericMeasure
, IfcParameterValue
, IfcPlaneAngleMeasure
, IfcPositiveLengthMeasure
, IfcPositivePlaneAngleMeasure
, IfcPositiveRatioMeasure
, IfcRatioMeasure
, IfcSolidAngleMeasure
, IfcThermodynamicTemperatureMeasure
, IfcTimeMeasure
, IfcVolumeMeasure);
END_TYPE;
```

```
TYPE IfcPresentationStyleSelect = SELECT
(IfcCurveStyle
, IfcNullStyle
, IfcSurfaceStyle);
END_TYPE;
```

```
TYPE IfcShell = SELECT
  (IfcClosedShell
   ,IfcOpenShell);
END_TYPE;
```

```
TYPE IfcSimpleValue = SELECT
  (IfcBoolean
   ,IfcIdentifier
   ,IfcInteger
   ,IfcLabel
   ,IfcLogical
   ,IfcReal
   ,IfcText);
END_TYPE;
```

```
TYPE IfcSizeSelect = SELECT
  (IfcDescriptiveMeasure
   ,IfcLengthMeasure
   ,IfcNormalisedRatioMeasure
   ,IfcPositiveLengthMeasure
   ,IfcPositiveRatioMeasure
   ,IfcRatioMeasure);
END_TYPE;
```

```
TYPE IfcSpecularHighlightSelect = SELECT
  (IfcSpecularExponent
   ,IfcSpecularRoughness);
END_TYPE;
```

```
TYPE IfcSurfaceStyleElementSelect = SELECT
  (IfcSurfaceStyleShading);
END_TYPE;
```

```
TYPE IfcTrimmingSelect = SELECT
  (IfcCartesianPoint
   ,IfcParameterValue);
END_TYPE;
```

```
TYPE IfcUnit = SELECT
  (IfcDerivedUnit
   ,IfcNamedUnit);
END_TYPE;
```

```
TYPE IfcValue = SELECT
  (IfcDerivedMeasureValue
   ,IfcMeasureValue
   ,IfcSimpleValue);
END_TYPE;
```

```
TYPE IfcVectorOrDirection = SELECT
  (IfcDirection
   ,IfcVector);
END_TYPE;
```

```
ENTITY IfcActorRole;
  Role : IfcRoleEnum;
```

```

    UserDefinedRole : OPTIONAL IfcLabel;
    Description : OPTIONAL IfcText;
WHERE
    WR1 : (Role <> IfcRoleEnum.USERDEFINED) OR
          ((Role = IfcRoleEnum.USERDEFINED) AND
           EXISTS(SELF.UserDefinedRole));
END_ENTITY;

ENTITY IfcActuatorType
  SUBTYPE OF (IfcDistributionControlElementType);
  PredefinedType : IfcActuatorTypeEnum;
END_ENTITY;

ENTITY IfcAddress
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcPostalAddress
     ,IfcTelecomAddress));
  Purpose : OPTIONAL IfcAddressTypeEnum;
  Description : OPTIONAL IfcText;
  UserDefinedPurpose : OPTIONAL IfcLabel;
  INVERSE
    OfPerson : SET [0:?] OF IfcPerson FOR Addresses;
    OfOrganization : SET [0:?] OF IfcOrganization FOR Addresses;
  WHERE
    WR1 : (NOT(EXISTS(Purpose))) OR
          ((Purpose <> IfcAddressTypeEnum.USERDEFINED) OR
           ((Purpose = IfcAddressTypeEnum.USERDEFINED) AND
            EXISTS(SELF.UserDefinedPurpose)));
END_ENTITY;

ENTITY IfcAirTerminalBoxType
  SUBTYPE OF (IfcFlowControllerType);
  PredefinedType : IfcAirTerminalBoxTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcAirTerminalBoxTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcAirTerminalBoxTypeEnum.USERDEFINED) AND
           EXISTS(SELF.IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcAirTerminalType
  SUBTYPE OF (IfcFlowTerminalType);
  PredefinedType : IfcAirTerminalTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcAirTerminalTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcAirTerminalTypeEnum.USERDEFINED) AND
           EXISTS(SELF.IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcAirToAirHeatRecoveryType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
  PredefinedType : IfcAirToAirHeatRecoveryTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcAirToAirHeatRecoveryTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcAirToAirHeatRecoveryTypeEnum.USERDEFINED) AND
           EXISTS(SELF.IfcElementType.ElementType));
END_ENTITY;

```

ENTITY IfcAlarmType

SUBTYPE OF (IfcDistributionControlElementType);

PredefinedType : IfcAlarmTypeEnum;

END_ENTITY;

ENTITY IfcApplication;

ApplicationDeveloper : IfcOrganization;

Version : IfcLabel;

ApplicationFullName : IfcLabel;

ApplicationIdentifier : IfcIdentifier;

UNIQUE

UR1 : ApplicationIdentifier;

UR2 : ApplicationFullName, Version;

END_ENTITY;

ENTITY IfcArbitraryClosedProfileDef

SUPERTYPE OF (ONEOF

(IfcArbitraryProfileDefWithVoids))

SUBTYPE OF (IfcProfileDef);

OuterCurve : IfcCurve;

WHERE

WR1 : OuterCurve.Dim = 2;

WR2 : NOT('IFC2X3_CV_V2_0.IFCLINE' IN TYPEOF(OuterCurve));

END_ENTITY;

ENTITY IfcArbitraryOpenProfileDef

SUPERTYPE OF (ONEOF

(IfcCenterLineProfileDef))

SUBTYPE OF (IfcProfileDef);

Curve : IfcBoundedCurve;

WHERE

WR11 : ('IFC2X3_CV_V2_0.IFCCENTERLINEPROFILEDEF' IN TYPEOF(SELF)) OR
(SELF\IfcProfileDef.ProfileType = IfcProfileTypeEnum.CURVE);

WR12 : Curve.Dim = 2;

END_ENTITY;

ENTITY IfcArbitraryProfileDefWithVoids

SUBTYPE OF (IfcArbitraryClosedProfileDef);

InnerCurves : SET [1:?] OF IfcCurve;

WHERE

WR1 : SELF\IfcProfileDef.ProfileType = AREA;

WR2 : SIZEOF(QUERY(temp <* InnerCurves | temp.Dim <> 2)) = 0;

WR3 : SIZEOF(QUERY(temp <* InnerCurves | 'IFC2X3_CV_V2_0.IFCLINE' IN TYPEOF(temp))) = 0;

END_ENTITY;

ENTITY IfcAsymmetricIShapeProfileDef

SUBTYPE OF (IfcIShapeProfileDef);

TopFlangeWidth : IfcPositiveLengthMeasure;

TopFlangeThickness : OPTIONAL IfcPositiveLengthMeasure;

TopFlangeFilletRadius : OPTIONAL IfcPositiveLengthMeasure;

CentreOfGravityInY : OPTIONAL IfcPositiveLengthMeasure;

END_ENTITY;

ENTITY IfcAxis1Placement

SUBTYPE OF (IfcPlacement);

Axis : OPTIONAL IfcDirection;

DERIVE

```

Z : IfcDirection := NVL (IfcNormalise(Axis), IfcRepresentationItem() ||
IfcGeometricRepresentationItem () || IfcDirection([0.0,0.0,1.0]));
WHERE
WR1 : (NOT (EXISTS (Axis))) OR (Axis.Dim = 3);
WR2 : SELF\IfcPlacement.Location.Dim = 3;
END_ENTITY;

```

```

ENTITY IfcAxis2Placement2D
SUBTYPE OF (IfcPlacement);
RefDirection : OPTIONAL IfcDirection;
DERIVE
P : LIST [2:2] OF IfcDirection := IfcBuild2Axes(RefDirection);
WHERE
WR1 : (NOT (EXISTS (RefDirection))) OR (RefDirection.Dim = 2);
WR2 : SELF\IfcPlacement.Location.Dim = 2;
END_ENTITY;

```

```

ENTITY IfcAxis2Placement3D
SUBTYPE OF (IfcPlacement);
Axis : OPTIONAL IfcDirection;
RefDirection : OPTIONAL IfcDirection;
DERIVE
P : LIST [3:3] OF IfcDirection := IfcBuildAxes(Axis, RefDirection);
WHERE
WR1 : SELF\IfcPlacement.Location.Dim = 3;
WR2 : (NOT (EXISTS (Axis))) OR (Axis.Dim = 3);
WR3 : (NOT (EXISTS (RefDirection))) OR (RefDirection.Dim = 3);
WR4 : (NOT (EXISTS (Axis))) OR (NOT (EXISTS (RefDirection))) OR
(IfcCrossProduct(Axis,RefDirection).Magnitude > 0.0);
WR5 : NOT ((EXISTS (Axis)) XOR (EXISTS (RefDirection)));
END_ENTITY;

```

```

ENTITY IfcBeam
SUBTYPE OF (IfcBuildingElement);
END_ENTITY;

```

```

ENTITY IfcBeamType
SUBTYPE OF (IfcBuildingElementType);
PredefinedType : IfcBeamTypeEnum;
END_ENTITY;

```

```

ENTITY IfcBoilerType
SUBTYPE OF (IfcEnergyConversionDeviceType);
PredefinedType : IfcBoilerTypeEnum;
WHERE
WR1 : (PredefinedType <> IfcBoilerTypeEnum.USERDEFINED) OR
((PredefinedType = IfcBoilerTypeEnum.USERDEFINED) AND
EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcBooleanClippingResult
SUBTYPE OF (IfcBooleanResult);
WHERE
WR1 : ('IFC2X3_CV_V2_0.IFCSWEPTAREASOLID' IN TYPEOF(FirstOperand)) OR
('IFC2X3_CV_V2_0.IFCBOOLEANCLIPPINGRESULT' IN TYPEOF(FirstOperand));
WR2 : ('IFC2X3_CV_V2_0.IFCHALFSPACESOLID' IN TYPEOF(SecondOperand));
WR3 : Operator = DIFFERENCE;

```

END_ENTITY;

```
ENTITY IfcBooleanResult
  SUPERTYPE OF (ONEOF
    (IfcBooleanClippingResult))
  SUBTYPE OF (IfcGeometricRepresentationItem);
    Operator : IfcBooleanOperator;
    FirstOperand : IfcBooleanOperand;
    SecondOperand : IfcBooleanOperand;
  DERIVE
    Dim : IfcDimensionCount := FirstOperand.Dim;
  WHERE
    WR1 : FirstOperand.Dim = SecondOperand.Dim;
END_ENTITY;
```

```
ENTITY IfcBoundedCurve
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcCompositeCurve
    ,IfcPolyline
    ,IfcTrimmedCurve))
  SUBTYPE OF (IfcCurve);
END_ENTITY;
```

```
ENTITY IfcBoundingBox
  SUBTYPE OF (IfcGeometricRepresentationItem);
    Corner : IfcCartesianPoint;
    XDim : IfcPositiveLengthMeasure;
    YDim : IfcPositiveLengthMeasure;
    ZDim : IfcPositiveLengthMeasure;
  DERIVE
    Dim : IfcDimensionCount := 3;
END_ENTITY;
```

```
ENTITY IfcBuilding
  SUBTYPE OF (IfcSpatialStructureElement);
    ElevationOfRefHeight : OPTIONAL IfcLengthMeasure;
    ElevationOfTerrain : OPTIONAL IfcLengthMeasure;
    BuildingAddress : OPTIONAL IfcPostalAddress;
END_ENTITY;
```

```
ENTITY IfcBuildingElement
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBeam
    ,IfcBuildingElementComponent
    ,IfcBuildingElementProxy
    ,IfcColumn
    ,IfcCovering
    ,IfcCurtainWall
    ,IfcDoor
    ,IfcFooting
    ,IfcMember
    ,IfcPile
    ,IfcPlate
    ,IfcRailing
    ,IfcRamp
    ,IfcRampFlight
    ,IfcRoof
```

```

    ,IfcSlab
    ,IfcStair
    ,IfcStairFlight
    ,IfcWall
    ,IfcWindow))
SUBTYPE OF (IfcElement);
END_ENTITY;

ENTITY IfcBuildingElementComponent
ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBuildingElementPart
    ,IfcReinforcingElement))
SUBTYPE OF (IfcBuildingElement);
END_ENTITY;

ENTITY IfcBuildingElementPart
SUBTYPE OF (IfcBuildingElementComponent);
END_ENTITY;

ENTITY IfcBuildingElementProxy
SUBTYPE OF (IfcBuildingElement);
    CompositionType : OPTIONAL IfcElementCompositionEnum;
WHERE
    WR1 : EXISTS(SELf\IfcRoot.Name);
END_ENTITY;

ENTITY IfcBuildingElementProxyType
SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : IfcBuildingElementProxyTypeEnum;
END_ENTITY;

ENTITY IfcBuildingElementType
ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBeamType
    ,IfcBuildingElementProxyType
    ,IfcColumnType
    ,IfcCoveringType
    ,IfcCurtainWallType
    ,IfcMemberType
    ,IfcPlateType
    ,IfcRailingType
    ,IfcRampFlightType
    ,IfcSlabType
    ,IfcStairFlightType
    ,IfcWallType))
SUBTYPE OF (IfcElementType);
END_ENTITY;

ENTITY IfcBuildingStorey
SUBTYPE OF (IfcSpatialStructureElement);
    Elevation : OPTIONAL IfcLengthMeasure;
END_ENTITY;

ENTITY IfcCShapeProfileDef
SUBTYPE OF (IfcParameterizedProfileDef);
    Depth : IfcPositiveLengthMeasure;
    Width : IfcPositiveLengthMeasure;

```

```

WallThickness : IfcPositiveLengthMeasure;
Girth : IfcPositiveLengthMeasure;
InternalFilletRadius : OPTIONAL IfcPositiveLengthMeasure;
CentreOfGravityInX : OPTIONAL IfcPositiveLengthMeasure;

```

```

WHERE

```

```

WR1 : Girth < (Depth / 2.);
WR2 : NOT(EXISTS(InternalFilletRadius)) OR
      ((InternalFilletRadius <= Width/2.) AND (InternalFilletRadius <= Depth/2.));
WR3 : (WallThickness < Width/2.) AND (WallThickness < Depth/2.);

```

```

END_ENTITY;

```

```

ENTITY IfcCableCarrierFittingType

```

```

  SUBTYPE OF (IfcFlowFittingType);

```

```

  PredefinedType : IfcCableCarrierFittingTypeEnum;

```

```

WHERE

```

```

WR1 : (PredefinedType <> IfcCableCarrierFittingTypeEnum.USERDEFINED) OR
      ((PredefinedType = IfcCableCarrierFittingTypeEnum.USERDEFINED) AND
       EXISTS(SELf\IfcElementType.ElementType));

```

```

END_ENTITY;

```

```

ENTITY IfcCableCarrierSegmentType

```

```

  SUBTYPE OF (IfcFlowSegmentType);

```

```

  PredefinedType : IfcCableCarrierSegmentTypeEnum;

```

```

END_ENTITY;

```

```

ENTITY IfcCableSegmentType

```

```

  SUBTYPE OF (IfcFlowSegmentType);

```

```

  PredefinedType : IfcCableSegmentTypeEnum;

```

```

END_ENTITY;

```

```

ENTITY IfcCartesianPoint

```

```

  SUBTYPE OF (IfcPoint);

```

```

  Coordinates : LIST [1:3] OF IfcLengthMeasure;

```

```

  DERIVE

```

```

    Dim : IfcDimensionCount := HIINDEX(Coordinates);

```

```

  WHERE

```

```

    WR1 : HIINDEX(Coordinates) >= 2;

```

```

END_ENTITY;

```

```

ENTITY IfcCartesianTransformationOperator

```

```

  ABSTRACT SUPERTYPE OF (ONEOF

```

```

    (IfcCartesianTransformationOperator2D
     ,IfcCartesianTransformationOperator3D))

```

```

  SUBTYPE OF (IfcGeometricRepresentationItem);

```

```

  Axis1 : OPTIONAL IfcDirection;

```

```

  Axis2 : OPTIONAL IfcDirection;

```

```

  LocalOrigin : IfcCartesianPoint;

```

```

  Scale : OPTIONAL REAL;

```

```

  DERIVE

```

```

    Sc1 : REAL := NVL(Scale, 1.0);

```

```

    Dim : IfcDimensionCount := LocalOrigin.Dim;

```

```

  WHERE

```

```

    WR1 : Sc1 > 0.0;

```

```

END_ENTITY;

```

```

ENTITY IfcCartesianTransformationOperator2D

```

```

  SUPERTYPE OF (ONEOF

```

```

    (IfcCartesianTransformationOperator2DnonUniform))
SUBTYPE OF (IfcCartesianTransformationOperator);
DERIVE
    U : LIST [2:2] OF IfcDirection :=
        IfcBaseAxis(2,SELF\IfcCartesianTransformationOperator.Axis1,
            SELF\IfcCartesianTransformationOperator.Axis2,?);
WHERE
    WR1 : SELF\IfcCartesianTransformationOperator.Dim = 2;
    WR2 : NOT(EXISTS(SELF\IfcCartesianTransformationOperator.Axis1)) OR
        (SELF\IfcCartesianTransformationOperator.Axis1.Dim = 2);
    WR3 : NOT(EXISTS(SELF\IfcCartesianTransformationOperator.Axis2)) OR
        (SELF\IfcCartesianTransformationOperator.Axis2.Dim = 2);
END_ENTITY;

ENTITY IfcCartesianTransformationOperator2DnonUniform
SUBTYPE OF (IfcCartesianTransformationOperator2D);
    Scale2 : OPTIONAL REAL;
DERIVE
    Sc12 : REAL := NVL(Scale2, SELF\IfcCartesianTransformationOperator.Scl);
WHERE
    WR1 : Sc12 > 0.0;
END_ENTITY;

ENTITY IfcCartesianTransformationOperator3D
SUPERTYPE OF (ONEOF
    (IfcCartesianTransformationOperator3DnonUniform))
SUBTYPE OF (IfcCartesianTransformationOperator);
    Axis3 : OPTIONAL IfcDirection;
DERIVE
    U : LIST [3:3] OF IfcDirection :=
        IfcBaseAxis(3,SELF\IfcCartesianTransformationOperator.Axis1,
            SELF\IfcCartesianTransformationOperator.Axis2,Axis3);
WHERE
    WR1 : SELF\IfcCartesianTransformationOperator.Dim = 3;
    WR2 : NOT(EXISTS(SELF\IfcCartesianTransformationOperator.Axis1)) OR
        (SELF\IfcCartesianTransformationOperator.Axis1.Dim = 3);
    WR3 : NOT(EXISTS(SELF\IfcCartesianTransformationOperator.Axis2)) OR
        (SELF\IfcCartesianTransformationOperator.Axis2.Dim = 3);
    WR4 : NOT(EXISTS(Axis3)) OR (Axis3.Dim = 3);
END_ENTITY;

ENTITY IfcCartesianTransformationOperator3DnonUniform
SUBTYPE OF (IfcCartesianTransformationOperator3D);
    Scale2 : OPTIONAL REAL;
    Scale3 : OPTIONAL REAL;
DERIVE
    Sc12 : REAL := NVL(Scale2, SELF\IfcCartesianTransformationOperator.Scl);
    Sc13 : REAL := NVL(Scale3, SELF\IfcCartesianTransformationOperator.Scl);
WHERE
    WR1 : Sc12 > 0.0;
    WR2 : Sc13 > 0.0;
END_ENTITY;

ENTITY IfcCenterLineProfileDef
SUBTYPE OF (IfcArbitraryOpenProfileDef);
    Thickness : IfcPositiveLengthMeasure;
END_ENTITY;

```

```

ENTITY IfcChillerType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcChillerTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcChillerTypeEnum.USERDEFINED) OR
      ((PredefinedType = IfcChillerTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcCircle
  SUBTYPE OF (IfcConic);
    Radius : IfcPositiveLengthMeasure;
END_ENTITY;

```

```

ENTITY IfcCircleHollowProfileDef
  SUBTYPE OF (IfcCircleProfileDef);
    WallThickness : IfcPositiveLengthMeasure;
  WHERE
    WR1 : WallThickness < SELF\IfcCircleProfileDef.Radius;
END_ENTITY;

```

```

ENTITY IfcCircleProfileDef
  SUPERTYPE OF (ONEOF
    (IfcCircleHollowProfileDef))
  SUBTYPE OF (IfcParameterizedProfileDef);
    Radius : IfcPositiveLengthMeasure;
END_ENTITY;

```

```

ENTITY IfcClassification;
  Source : IfcLabel;
  Edition : IfcLabel;
  EditionDate : OPTIONAL IfcStrippedOptional;
  Name : IfcLabel;
END_ENTITY;

```

```

ENTITY IfcClassificationReference
  SUBTYPE OF (IfcExternalReference);
    ReferencedSource : OPTIONAL IfcClassification;
END_ENTITY;

```

```

ENTITY IfcClosedShell
  SUBTYPE OF (IfcConnectedFaceSet);
END_ENTITY;

```

```

ENTITY IfcCoilType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcCoilTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcCoilTypeEnum.USERDEFINED) OR
      ((PredefinedType = IfcCoilTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcColourRgb
  SUBTYPE OF (IfcColourSpecification);
    Red : IfcNormalisedRatioMeasure;

```

```

    Green : IfcNormalisedRatioMeasure;
    Blue  : IfcNormalisedRatioMeasure;
END_ENTITY;

```

```

ENTITY IfcColourSpecification
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcColourRgb));
    Name : OPTIONAL IfcLabel;
END_ENTITY;

```

```

ENTITY IfcColumn
  SUBTYPE OF (IfcBuildingElement);
END_ENTITY;

```

```

ENTITY IfcColumnType
  SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : IfcColumnTypeEnum;
END_ENTITY;

```

```

ENTITY IfcComplexProperty
  SUBTYPE OF (IfcProperty);
    UsageName : IfcIdentifier;
    HasProperties : SET [1:?] OF IfcProperty;
  WHERE
    WR21 : SIZEOF(QUERY(temp <* HasProperties | SELF :=: temp)) = 0;
    WR22 : IfcUniquePropertyName(HasProperties);
END_ENTITY;

```

```

ENTITY IfcCompositeCurve
  SUBTYPE OF (IfcBoundedCurve);
    Segments : LIST [1:?] OF IfcCompositeCurveSegment;
    SelfIntersect : LOGICAL;
  DERIVE
    NSegments : INTEGER := SIZEOF(Segments);
    ClosedCurve : LOGICAL := Segments[NSegments].Transition <> Discontinuous;
  WHERE
    WR41 : ((NOT ClosedCurve) AND (SIZEOF(QUERY(Temp <* Segments | Temp.Transition =
      Discontinuous)) = 1)) OR ((ClosedCurve) AND (SIZEOF(QUERY(Temp <* Segments |
      Temp.Transition = Discontinuous)) = 0));
    WR42 : SIZEOF( QUERY( Temp <* Segments | Temp.Dim <> Segments[1].Dim)) = 0;
END_ENTITY;

```

```

ENTITY IfcCompositeCurveSegment
  SUBTYPE OF (IfcGeometricRepresentationItem);
    Transition : IfcTransitionCode;
    SameSense : BOOLEAN;
    ParentCurve : IfcCurve;
  DERIVE
    Dim : IfcDimensionCount := ParentCurve.Dim;
  INVERSE
    UsingCurves : SET [1:?] OF IfcCompositeCurve FOR Segments;
  WHERE
    WR1 : ('IFC2X3_CV_V2_0.IFCBOUNDEDCURVE' IN TYPEOF(ParentCurve));
END_ENTITY;

```

```

ENTITY IfcCompositeProfileDef
  SUBTYPE OF (IfcProfileDef);

```

```

    Profiles : SET [2:?] OF IfcProfileDef;
    Label : OPTIONAL IfcLabel;
WHERE
    WR1 : SIZEOF(QUERY(temp <* Profiles | temp.ProfileType <> Profiles[1].ProfileType)) = 0;
    WR2 : SIZEOF(QUERY(temp <* Profiles | 'IFC2X3_CV_V2_0.IFCCOMPOSITEPROFILEDEF' IN
        TYPEOF(temp))) = 0;
END_ENTITY;

ENTITY IfcCompressorType
    SUBTYPE OF (IfcFlowMovingDeviceType);
    PredefinedType : IfcCompressorTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcCompressorTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcCompressorTypeEnum.USERDEFINED) AND
        EXISTS(SELf\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcCondenserType
    SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcCondenserTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcCondenserTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcCondenserTypeEnum.USERDEFINED) AND
        EXISTS(SELf\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcConic
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcCircle
        ,IfcEllipse))
    SUBTYPE OF (IfcCurve);
    Position : IfcAxis2Placement;
END_ENTITY;

ENTITY IfcConnectedFaceSet
    SUPERTYPE OF (ONEOF
        (IfcClosedShell
        ,IfcOpenShell))
    SUBTYPE OF (IfcTopologicalRepresentationItem);
    CfsFaces : SET [1:?] OF IfcFace;
END_ENTITY;

ENTITY IfcConnectionCurveGeometry
    SUBTYPE OF (IfcConnectionGeometry);
    CurveOnRelatingElement : IfcCurveOrEdgeCurve;
    CurveOnRelatedElement : OPTIONAL IfcCurveOrEdgeCurve;
END_ENTITY;

ENTITY IfcConnectionGeometry
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcConnectionCurveGeometry));
END_ENTITY;

ENTITY IfcControllerType
    SUBTYPE OF (IfcDistributionControlElementType);
    PredefinedType : IfcControllerTypeEnum;
END_ENTITY;

```

ENTITY IfcConversionBasedUnit

SUBTYPE OF (IfcNamedUnit);

 Name : IfcLabel;

 ConversionFactor : IfcMeasureWithUnit;

END_ENTITY;

ENTITY IfcCooledBeamType

SUBTYPE OF (IfcEnergyConversionDeviceType);

 PredefinedType : IfcCooledBeamTypeEnum;

WHERE

 WR1 : (PredefinedType <> IfcCooledBeamTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcCooledBeamTypeEnum.USERDEFINED) AND
 EXISTS(SELF\IfcElementType.ElementType));

END_ENTITY;

ENTITY IfcCoolingTowerType

SUBTYPE OF (IfcEnergyConversionDeviceType);

 PredefinedType : IfcCoolingTowerTypeEnum;

WHERE

 WR1 : (PredefinedType <> IfcCoolingTowerTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcCoolingTowerTypeEnum.USERDEFINED) AND
 EXISTS(SELF\IfcElementType.ElementType));

END_ENTITY;

ENTITY IfcCovering

SUBTYPE OF (IfcBuildingElement);

 PredefinedType : OPTIONAL IfcCoveringTypeEnum;

WHERE

 WR61 : NOT(EXISTS(PredefinedType)) OR
 (PredefinedType <> IfcCoveringTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcCoveringTypeEnum.USERDEFINED) AND EXISTS
 (SELF\IfcObject.ObjectType));

END_ENTITY;

ENTITY IfcCoveringType

SUBTYPE OF (IfcBuildingElementType);

 PredefinedType : IfcCoveringTypeEnum;

END_ENTITY;

ENTITY IfcCurtainWall

SUBTYPE OF (IfcBuildingElement);

END_ENTITY;

ENTITY IfcCurtainWallType

SUBTYPE OF (IfcBuildingElementType);

 PredefinedType : IfcCurtainWallTypeEnum;

END_ENTITY;

ENTITY IfcCurve

ABSTRACT SUPERTYPE OF (ONEOF

 (IfcBoundedCurve

 , IfcConic

 , IfcLine))

SUBTYPE OF (IfcGeometricRepresentationItem);

DERIVE

 Dim : IfcDimensionCount := IfcCurveDim(SELF);

END_ENTITY;

ENTITY IfcCurveStyle

SUBTYPE OF (IfcPresentationStyle);

CurveFont : OPTIONAL IfcStrippedOptional;

CurveWidth : OPTIONAL IfcSizeSelect;

CurveColour : OPTIONAL IfcColour;

WHERE

WR1 : (NOT(EXISTS(CurveWidth))) OR
 ('IFC2X3_CV_V2_0.IFCPOSITIVELENGTHMEASURE' IN TYPEOF(CurveWidth)) OR
 (('IFC2X3_CV_V2_0.IFCDESCRIPTIVEMEASURE' IN TYPEOF(CurveWidth)) AND
 (CurveWidth = 'by layer'));

END_ENTITY;

ENTITY IfcDamperType

SUBTYPE OF (IfcFlowControllerType);

PredefinedType : IfcDamperTypeEnum;

WHERE

WR1 : (PredefinedType <> IfcDamperTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcDamperTypeEnum.USERDEFINED) AND
 EXISTS(SELf\IfcElementType.ElementType));

END_ENTITY;

ENTITY IfcDerivedProfileDef

SUBTYPE OF (IfcProfileDef);

ParentProfile : IfcProfileDef;

Operator : IfcCartesianTransformationOperator2D;

Label : OPTIONAL IfcLabel;

WHERE

WR1 : SELF\IfcProfileDef.ProfileType = ParentProfile.ProfileType;

END_ENTITY;

ENTITY IfcDerivedUnit;

Elements : SET [1:?] OF IfcDerivedUnitElement;

UnitType : IfcDerivedUnitEnum;

UserDefinedType : OPTIONAL IfcLabel;

DERIVE

Dimensions : IfcDimensionalExponents := IfcDeriveDimensionalExponents(Elements);

WHERE

WR1 : (SIZEOF (Elements) > 1) OR ((SIZEOF (Elements) = 1) AND (Elements[1].Exponent <> 1
));
 WR2 : (UnitType <> IfcDerivedUnitEnum.USERDEFINED) OR
 ((UnitType = IfcDerivedUnitEnum.USERDEFINED) AND
 (EXISTS(SELf.UserDefinedType)));

END_ENTITY;

ENTITY IfcDerivedUnitElement;

Unit : IfcNamedUnit;

Exponent : INTEGER;

END_ENTITY;

ENTITY IfcDimensionalExponents;

LengthExponent : INTEGER;

MassExponent : INTEGER;

TimeExponent : INTEGER;

ElectricCurrentExponent : INTEGER;

ThermodynamicTemperatureExponent : INTEGER;

```

    AmountOfSubstanceExponent : INTEGER;
    LuminousIntensityExponent : INTEGER;
END_ENTITY;

ENTITY IfcDirection
  SUBTYPE OF (IfcGeometricRepresentationItem);
    DirectionRatios : LIST [2:3] OF REAL;
  DERIVE
    Dim : IfcDimensionCount := HIINDEX(DirectionRatios);
END_ENTITY;

ENTITY IfcDiscreteAccessory
  SUBTYPE OF (IfcElementComponent);
END_ENTITY;

ENTITY IfcDiscreteAccessoryType
  SUBTYPE OF (IfcElementComponentType);
END_ENTITY;

ENTITY IfcDistributionChamberElement
  SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

ENTITY IfcDistributionChamberElementType
  SUBTYPE OF (IfcDistributionFlowElementType);
    PredefinedType : IfcDistributionChamberElementTypeEnum;
END_ENTITY;

ENTITY IfcDistributionControlElement
  SUBTYPE OF (IfcDistributionElement);
    ControlElementId : OPTIONAL IfcIdentifier;
  INVERSE
    AssignedToFlowElement : SET [0:1] OF IfcRelFlowControlElements FOR RelatedControlElements;
END_ENTITY;

ENTITY IfcDistributionControlElementType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcActuatorType
    ,IfcAlarmType
    ,IfcControllerType
    ,IfcFlowInstrumentType
    ,IfcSensorType))
  SUBTYPE OF (IfcDistributionElementType);
END_ENTITY;

ENTITY IfcDistributionElement
  SUPERTYPE OF (ONEOF
    (IfcDistributionControlElement
    ,IfcDistributionFlowElement))
  SUBTYPE OF (IfcElement);
END_ENTITY;

ENTITY IfcDistributionElementType
  SUPERTYPE OF (ONEOF
    (IfcDistributionControlElementType
    ,IfcDistributionFlowElementType))
  SUBTYPE OF (IfcElementType);

```

END_ENTITY;

ENTITY IfcDistributionFlowElement

SUPERTYPE OF (ONEOF

(IfcDistributionChamberElement
, IfcEnergyConversionDevice
, IfcFlowController
, IfcFlowFitting
, IfcFlowMovingDevice
, IfcFlowSegment
, IfcFlowStorageDevice
, IfcFlowTerminal
, IfcFlowTreatmentDevice))

SUBTYPE OF (IfcDistributionElement);

INVERSE

HasControlElements : SET [0:1] OF IfcRelFlowControlElements FOR RelatingFlowElement;

END_ENTITY;

ENTITY IfcDistributionFlowElementType

ABSTRACT SUPERTYPE OF (ONEOF

(IfcDistributionChamberElementType
, IfcEnergyConversionDeviceType
, IfcFlowControllerType
, IfcFlowFittingType
, IfcFlowMovingDeviceType
, IfcFlowSegmentType
, IfcFlowStorageDeviceType
, IfcFlowTerminalType
, IfcFlowTreatmentDeviceType))

SUBTYPE OF (IfcDistributionElementType);

END_ENTITY;

ENTITY IfcDistributionPort

SUBTYPE OF (IfcPort);

FlowDirection : OPTIONAL IfcFlowDirectionEnum;

END_ENTITY;

ENTITY IfcDoor

SUBTYPE OF (IfcBuildingElement);

OverallHeight : OPTIONAL IfcPositiveLengthMeasure;

OverallWidth : OPTIONAL IfcPositiveLengthMeasure;

END_ENTITY;

ENTITY IfcDoorLiningProperties

SUBTYPE OF (IfcPropertySetDefinition);

LiningDepth : OPTIONAL IfcPositiveLengthMeasure;

LiningThickness : OPTIONAL IfcPositiveLengthMeasure;

ThresholdDepth : OPTIONAL IfcPositiveLengthMeasure;

ThresholdThickness : OPTIONAL IfcPositiveLengthMeasure;

TransomThickness : OPTIONAL IfcPositiveLengthMeasure;

TransomOffset : OPTIONAL IfcLengthMeasure;

LiningOffset : OPTIONAL IfcLengthMeasure;

ThresholdOffset : OPTIONAL IfcLengthMeasure;

CasingThickness : OPTIONAL IfcPositiveLengthMeasure;

CasingDepth : OPTIONAL IfcPositiveLengthMeasure;

ShapeAspectStyle : OPTIONAL IfcStrippedOptional;

WHERE

```

WR31 : NOT(NOT(EXISTS(LiningDepth)) AND EXISTS(LiningThickness));
WR32 : NOT(NOT(EXISTS(ThresholdDepth)) AND EXISTS(ThresholdThickness));
WR33 : (EXISTS(TransomOffset) AND EXISTS(TransomThickness)) XOR
        (NOT(EXISTS(TransomOffset)) AND NOT(EXISTS(TransomThickness)));
WR34 : (EXISTS(CasingDepth) AND EXISTS(CasingThickness)) XOR
        (NOT(EXISTS(CasingDepth)) AND NOT(EXISTS(CasingThickness)));
WR35 : EXISTS(SELf\IfcPropertySetDefinition.DefinesType[1]) AND
        ('IFC2X3_CV_V2_0.IFCD00RSTYLE' IN
        TYPEOF(SELf\IfcPropertySetDefinition.DefinesType[1]));

```

END_ENTITY;

ENTITY IfcDoorPanelProperties

```

SUBTYPE OF (IfcPropertySetDefinition);
PanelDepth : OPTIONAL IfcPositiveLengthMeasure;
PanelOperation : IfcDoorPanelOperationEnum;
PanelWidth : OPTIONAL IfcNormalisedRatioMeasure;
PanelPosition : IfcDoorPanelPositionEnum;
ShapeAspectStyle : OPTIONAL IfcStrippedOptional;

```

WHERE

```

WR31 : EXISTS(SELf\IfcPropertySetDefinition.DefinesType[1]) AND
        ('IFC2X3_CV_V2_0.IFCD00RSTYLE' IN
        TYPEOF(SELf\IfcPropertySetDefinition.DefinesType[1]));

```

END_ENTITY;

ENTITY IfcDoorStyle

```

SUBTYPE OF (IfcTypeProduct);
OperationType : IfcDoorStyleOperationEnum;
ConstructionType : IfcDoorStyleConstructionEnum;
ParameterTakesPrecedence : BOOLEAN;
Sizeable : BOOLEAN;

```

END_ENTITY;

ENTITY IfcDuctFittingType

```

SUBTYPE OF (IfcFlowFittingType);
PredefinedType : IfcDuctFittingTypeEnum;

```

WHERE

```

WR2 : (PredefinedType <> IfcDuctFittingTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcDuctFittingTypeEnum.USERDEFINED) AND
        EXISTS(SELf\IfcElementType.ElementType));

```

END_ENTITY;

ENTITY IfcDuctSegmentType

```

SUBTYPE OF (IfcFlowSegmentType);
PredefinedType : IfcDuctSegmentTypeEnum;

```

WHERE

```

WR1 : (PredefinedType <> IfcDuctSegmentTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcDuctSegmentTypeEnum.USERDEFINED) AND
        EXISTS(SELf\IfcElementType.ElementType));

```

END_ENTITY;

ENTITY IfcDuctSilencerType

```

SUBTYPE OF (IfcFlowTreatmentDeviceType);
PredefinedType : IfcDuctSilencerTypeEnum;

```

WHERE

```

WR1 : (PredefinedType <> IfcDuctSilencerTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcDuctSilencerTypeEnum.USERDEFINED) AND
        EXISTS(SELf\IfcElementType.ElementType));

```

END_ENTITY;

```
ENTITY IfcElectricApplianceType
  SUBTYPE OF (IfcFlowTerminalType);
    PredefinedType : IfcElectricApplianceTypeEnum;
END_ENTITY;
```

```
ENTITY IfcElectricFlowStorageDeviceType
  SUBTYPE OF (IfcFlowStorageDeviceType);
    PredefinedType : IfcElectricFlowStorageDeviceTypeEnum;
END_ENTITY;
```

```
ENTITY IfcElectricGeneratorType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcElectricGeneratorTypeEnum;
END_ENTITY;
```

```
ENTITY IfcElectricMotorType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcElectricMotorTypeEnum;
END_ENTITY;
```

```
ENTITY IfcElectricTimeControlType
  SUBTYPE OF (IfcFlowControllerType);
    PredefinedType : IfcElectricTimeControlTypeEnum;
END_ENTITY;
```

```
ENTITY IfcElement
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBuildingElement
    ,IfcDistributionElement
    ,IfcElementAssembly
    ,IfcElementComponent
    ,IfcFeatureElement
    ,IfcFurnishingElement
    ,IfcTransportElement))
  SUBTYPE OF (IfcProduct);
    Tag : OPTIONAL IfcIdentifier;
  INVERSE
    FillsVoids : SET [0:1] OF IfcRelFillsElement FOR RelatedBuildingElement;
    ConnectedTo : SET [0:?] OF IfcRelConnectsElements FOR RelatingElement;
    HasPorts : SET [0:?] OF IfcRelConnectsPortToElement FOR RelatedElement;
    HasOpenings : SET [0:?] OF IfcRelVoidsElement FOR RelatingBuildingElement;
    IsConnectionRealization : SET [0:?] OF IfcRelConnectsWithRealizingElements FOR
      RealizingElements;
    ConnectedFrom : SET [0:?] OF IfcRelConnectsElements FOR RelatedElement;
    ContainedInStructure : SET [0:1] OF IfcRelContainedInSpatialStructure FOR RelatedElements;
END_ENTITY;
```

```
ENTITY IfcElementAssembly
  SUBTYPE OF (IfcElement);
    AssemblyPlace : OPTIONAL IfcAssemblyPlaceEnum;
    PredefinedType : IfcElementAssemblyTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcElementAssemblyTypeEnum.USERDEFINED) OR ((PredefinedType =
      IfcElementAssemblyTypeEnum.USERDEFINED) AND EXISTS(SELF\IfcObject.ObjectType));
END_ENTITY;
```

```
ENTITY IfcElementComponent
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcDiscreteAccessory
    ,IfcFastener))
  SUBTYPE OF (IfcElement);
END_ENTITY;
```

```
ENTITY IfcElementComponentType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcDiscreteAccessoryType
    ,IfcFastenerType))
  SUBTYPE OF (IfcElementType);
END_ENTITY;
```

```
ENTITY IfcElementType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBuildingElementType
    ,IfcDistributionElementType
    ,IfcElementComponentType
    ,IfcFurnishingElementType
    ,IfcSpatialStructureElementType
    ,IfcTransportElementType))
  SUBTYPE OF (IfcTypeProduct);
  ElementType : OPTIONAL IfcLabel;
END_ENTITY;
```

```
ENTITY IfcElementarySurface
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcPlane))
  SUBTYPE OF (IfcSurface);
  Position : IfcAxis2Placement3D;
  DERIVE
    Dim : IfcDimensionCount := Position.Dim;
END_ENTITY;
```

```
ENTITY IfcEllipse
  SUBTYPE OF (IfcConic);
  SemiAxis1 : IfcPositiveLengthMeasure;
  SemiAxis2 : IfcPositiveLengthMeasure;
END_ENTITY;
```

```
ENTITY IfcEllipseProfileDef
  SUBTYPE OF (IfcParameterizedProfileDef);
  SemiAxis1 : IfcPositiveLengthMeasure;
  SemiAxis2 : IfcPositiveLengthMeasure;
END_ENTITY;
```

```
ENTITY IfcEnergyConversionDevice
  SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;
```

```
ENTITY IfcEnergyConversionDeviceType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcAirToAirHeatRecoveryType
    ,IfcBoilerType
    ,IfcChillerType
```

```

, IfcCoilType
, IfcCondenserType
, IfcCooledBeamType
, IfcCoolingTowerType
, IfcElectricGeneratorType
, IfcElectricMotorType
, IfcEvaporativeCoolerType
, IfcEvaporatorType
, IfcHeatExchangerType
, IfcHumidifierType
, IfcMotorConnectionType
, IfcSpaceHeaterType
, IfcTransformerType
, IfcTubeBundleType
, IfcUnitaryEquipmentType))
SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

ENTITY IfcEvaporativeCoolerType
SUBTYPE OF (IfcEnergyConversionDeviceType);
  PredefinedType : IfcEvaporativeCoolerTypeEnum;
WHERE
  WR1 : (PredefinedType <> IfcEvaporativeCoolerTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcEvaporativeCoolerTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcEvaporatorType
SUBTYPE OF (IfcEnergyConversionDeviceType);
  PredefinedType : IfcEvaporatorTypeEnum;
WHERE
  WR1 : (PredefinedType <> IfcEvaporatorTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcEvaporatorTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcExternalReference
ABSTRACT SUPERTYPE OF (ONEOF
  (IfcClassificationReference));
  Location : OPTIONAL IfcLabel;
  ItemReference : OPTIONAL IfcIdentifier;
  Name : OPTIONAL IfcLabel;
WHERE
  WR1 : EXISTS(ItemReference) OR EXISTS(Location) OR EXISTS(Name);
END_ENTITY;

ENTITY IfcExtrudedAreaSolid
SUBTYPE OF (IfcSweptAreaSolid);
  ExtrudedDirection : IfcDirection;
  Depth : IfcPositiveLengthMeasure;
WHERE
  WR31 : IfcDotProduct(IfcRepresentationItem() || IfcGeometricRepresentationItem() ||
    IfcDirection([0.0,0.0,1.0]), SELF.ExtrudedDirection) <> 0.0;
END_ENTITY;

ENTITY IfcFace
SUBTYPE OF (IfcTopologicalRepresentationItem);

```

```

    Bounds : SET [1:?] OF IfcFaceBound;
WHERE
    WR1 : SIZEOF(QUERY(temp <* Bounds | 'IFC2X3_CV_V2_0.IFCFACEOUTERBOUND' IN TYPEOF(temp)))
    <= 1;
END_ENTITY;

ENTITY IfcFaceBasedSurfaceModel
    SUBTYPE OF (IfcGeometricRepresentationItem);
    FbsmFaces : SET [1:?] OF IfcConnectedFaceSet;
    DERIVE
        Dim : IfcDimensionCount := 3;
END_ENTITY;

ENTITY IfcFaceBound
    SUPERTYPE OF (ONEOF
        (IfcFaceOuterBound))
    SUBTYPE OF (IfcTopologicalRepresentationItem);
    Bound : IfcLoop;
    Orientation : BOOLEAN;
END_ENTITY;

ENTITY IfcFaceOuterBound
    SUBTYPE OF (IfcFaceBound);
END_ENTITY;

ENTITY IfcFacetedBrep
    SUBTYPE OF (IfcManifoldSolidBrep);
END_ENTITY;

ENTITY IfcFanType
    SUBTYPE OF (IfcFlowMovingDeviceType);
    PredefinedType : IfcFanTypeEnum;
    WHERE
        WR1 : (PredefinedType <> IfcFanTypeEnum.USERDEFINED) OR
            ((PredefinedType = IfcFanTypeEnum.USERDEFINED) AND
            EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcFastener
    SUPERTYPE OF (ONEOF
        (IfcMechanicalFastener))
    SUBTYPE OF (IfcElementComponent);
END_ENTITY;

ENTITY IfcFastenerType
    SUPERTYPE OF (ONEOF
        (IfcMechanicalFastenerType))
    SUBTYPE OF (IfcElementComponentType);
END_ENTITY;

ENTITY IfcFeatureElement
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcFeatureElementSubtraction))
    SUBTYPE OF (IfcElement);
END_ENTITY;

ENTITY IfcFeatureElementSubtraction

```

```

ABSTRACT SUPERTYPE OF (ONEOF
  (IfcOpeningElement))
SUBTYPE OF (IfcFeatureElement);
INVERSE
  VoidsElements : IfcRelVoidsElement FOR RelatedOpeningElement;
END_ENTITY;

```

```

ENTITY IfcFilterType
  SUBTYPE OF (IfcFlowTreatmentDeviceType);
  PredefinedType : IfcFilterTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcFilterTypeEnum.USERDEFINED) OR
      ((PredefinedType = IfcFilterTypeEnum.USERDEFINED) AND
        EXISTS(SELFI\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcFireSuppressionTerminalType
  SUBTYPE OF (IfcFlowTerminalType);
  PredefinedType : IfcFireSuppressionTerminalTypeEnum;
END_ENTITY;

```

```

ENTITY IfcFlowController
  SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowControllerType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcAirTerminalBoxType
    ,IfcDamperType
    ,IfcElectricTimeControlType
    ,IfcFlowMeterType
    ,IfcProtectiveDeviceType
    ,IfcSwitchingDeviceType
    ,IfcValveType))
  SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowFitting
  SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowFittingType
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcCableCarrierFittingType
    ,IfcDuctFittingType
    ,IfcJunctionBoxType
    ,IfcPipeFittingType))
  SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowInstrumentType
  SUBTYPE OF (IfcDistributionControlElementType);
  PredefinedType : IfcFlowInstrumentTypeEnum;
END_ENTITY;

```

```

ENTITY IfcFlowMeterType
  SUBTYPE OF (IfcFlowControllerType);

```

```

    PredefinedType : IfcFlowMeterTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcFlowMeterTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcFlowMeterTypeEnum.USERDEFINED) AND
           EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcFlowMovingDevice
    SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowMovingDeviceType
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcCompressorType
         ,IfcFanType
         ,IfcPumpType))
    SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowSegment
    SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowSegmentType
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcCableCarrierSegmentType
         ,IfcCableSegmentType
         ,IfcDuctSegmentType
         ,IfcPipeSegmentType))
    SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowStorageDevice
    SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowStorageDeviceType
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcElectricFlowStorageDeviceType
         ,IfcTankType))
    SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowTerminal
    SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowTerminalType
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcAirTerminalType
         ,IfcElectricApplianceType
         ,IfcFireSuppressionTerminalType
         ,IfcLampType
         ,IfcLightFixtureType
         ,IfcOutletType
         ,IfcSanitaryTerminalType
         ,IfcStackTerminalType

```

```

    ,IfcWasteTerminalType))
SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFlowTreatmentDevice
SUBTYPE OF (IfcDistributionFlowElement);
END_ENTITY;

```

```

ENTITY IfcFlowTreatmentDeviceType
ABSTRACT SUPERTYPE OF (ONEOF
    (IfcDuctSilencerType
    ,IfcFilterType))
SUBTYPE OF (IfcDistributionFlowElementType);
END_ENTITY;

```

```

ENTITY IfcFooting
SUBTYPE OF (IfcBuildingElement);
    PredefinedType : IfcFootingTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcFootingTypeEnum.USERDEFINED) OR ((PredefinedType =
        IfcFootingTypeEnum.USERDEFINED) AND EXISTS(SELF\IfcObject.ObjectType));
END_ENTITY;

```

```

ENTITY IfcFurnishingElement
SUBTYPE OF (IfcElement);
END_ENTITY;

```

```

ENTITY IfcFurnishingElementType
SUBTYPE OF (IfcElementType);
END_ENTITY;

```

```

ENTITY IfcGeometricCurveSet
SUBTYPE OF (IfcGeometricSet);
WHERE
    WR1 : SIZEOF(QUERY(Temp <* SELF\IfcGeometricSet.Elements |
        'IFC2X3_CV_V2_0.IFCSURFACE' IN TYPEOF(Temp))) = 0;
END_ENTITY;

```

```

ENTITY IfcGeometricRepresentationContext
SUPERTYPE OF (ONEOF
    (IfcGeometricRepresentationSubContext))
SUBTYPE OF (IfcRepresentationContext);
    CoordinateSpaceDimension : IfcDimensionCount;
    Precision : OPTIONAL REAL;
    WorldCoordinateSystem : IfcAxis2Placement;
    TrueNorth : OPTIONAL IfcDirection;
INVERSE
    HasSubContexts : SET [0:?] OF IfcGeometricRepresentationSubContext FOR ParentContext;
END_ENTITY;

```

```

ENTITY IfcGeometricRepresentationItem
ABSTRACT SUPERTYPE OF (ONEOF
    (IfcBooleanResult
    ,IfcBoundingBox
    ,IfcCartesianTransformationOperator
    ,IfcCompositeCurveSegment
    ,IfcCurve

```

```

,IfcDirection
,IfcFaceBasedSurfaceModel
,IfcGeometricSet
,IfcHalfSpaceSolid
,IfcPlacement
,IfcPoint
,IfcShellBasedSurfaceModel
,IfcSolidModel
,IfcSurface
,IfcVector))
SUBTYPE OF (IfcRepresentationItem);
END_ENTITY;

ENTITY IfcGeometricRepresentationSubContext
SUBTYPE OF (IfcGeometricRepresentationContext);
  ParentContext : IfcGeometricRepresentationContext;
  TargetScale : OPTIONAL IfcPositiveRatioMeasure;
  TargetView : IfcGeometricProjectionEnum;
  UserDefinedTargetView : OPTIONAL IfcLabel;
DERIVE
  SELF\IfcGeometricRepresentationContext.WorldCoordinateSystem : IfcAxis2Placement :=
  ParentContext.WorldCoordinateSystem;
  SELF\IfcGeometricRepresentationContext.CoordinateSpaceDimension : IfcDimensionCount :=
  ParentContext.CoordinateSpaceDimension;
  SELF\IfcGeometricRepresentationContext.TrueNorth : IfcDirection :=
  NVL(ParentContext.TrueNorth,SELF.WorldCoordinateSystem.P[2]);
  SELF\IfcGeometricRepresentationContext.Precision : REAL :=
  NVL(ParentContext.Precision,1.E-5);
WHERE
  WR31 : NOT('IFC2X3_CV_V2_0.IFCGEOMETRICREPRESENTATIONSUBCONTEXT' IN
  TYPEOF(ParentContext));
  WR32 : (TargetView <> IfcGeometricProjectionEnum.USERDEFINED) OR
  ((TargetView = IfcGeometricProjectionEnum.USERDEFINED) AND
  EXISTS(UserDefinedTargetView));
END_ENTITY;

ENTITY IfcGeometricSet
SUPERTYPE OF (ONEOF
  (IfcGeometricCurveSet))
SUBTYPE OF (IfcGeometricRepresentationItem);
  Elements : SET [1:?] OF IfcGeometricSetSelect;
DERIVE
  Dim : IfcDimensionCount := Elements[1].Dim;
WHERE
  WR21 : SIZEOF(QUERY(Temp <* Elements |
  Temp.Dim <> Elements[1].Dim))
  = 0;
END_ENTITY;

ENTITY IfcGrid
SUBTYPE OF (IfcProduct);
  UAxes : LIST [1:?] OF UNIQUE IfcGridAxis;
  VAxes : LIST [1:?] OF UNIQUE IfcGridAxis;
  WAxes : OPTIONAL LIST [1:?] OF UNIQUE IfcGridAxis;
INVERSE
  ContainedInStructure : SET [0:1] OF IfcRelContainedInSpatialStructure FOR RelatedElements;
WHERE

```

```

    WR41 : EXISTS(SELF\IfcProduct.ObjectPlacement);
END_ENTITY;

```

```

ENTITY IfcGridAxis;
    AxisTag : OPTIONAL IfcLabel;
    AxisCurve : IfcCurve;
    SameSense : IfcBoolean;
INVERSE
    PartOfW : SET [0:1] OF IfcGrid FOR WAxes;
    PartOfV : SET [0:1] OF IfcGrid FOR VAxes;
    PartOfU : SET [0:1] OF IfcGrid FOR UAxes;
WHERE
    WR1 : AxisCurve.Dim = 2;
    WR2 : (SIZEOF(PartOfU) = 1) XOR (SIZEOF(PartOfV) = 1) XOR (SIZEOF(PartOfW) = 1);
END_ENTITY;

```

```

ENTITY IfcGroup
    SUPERTYPE OF (ONEOF
        (IfcSystem
        ,IfcZone))
    SUBTYPE OF (IfcObject);
INVERSE
    IsGroupedBy : IfcRelAssignsToGroup FOR RelatingGroup;
END_ENTITY;

```

```

ENTITY IfcHalfSpaceSolid
    SUPERTYPE OF (ONEOF
        (IfcPolygonalBoundedHalfSpace))
    SUBTYPE OF (IfcGeometricRepresentationItem);
    BaseSurface : IfcSurface;
    AgreementFlag : BOOLEAN;
DERIVE
    Dim : IfcDimensionCount := 3;
END_ENTITY;

```

```

ENTITY IfcHeatExchangerType
    SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcHeatExchangerTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcHeatExchangerTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcHeatExchangerTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcHumidifierType
    SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcHumidifierTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcHumidifierTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcHumidifierTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcIShapeProfileDef
    SUPERTYPE OF (ONEOF
        (IfcAsymmetricIShapeProfileDef))
    SUBTYPE OF (IfcParameterizedProfileDef);

```

```

OverallWidth : IfcPositiveLengthMeasure;
OverallDepth : IfcPositiveLengthMeasure;
WebThickness : IfcPositiveLengthMeasure;
FlangeThickness : IfcPositiveLengthMeasure;
FilletRadius : OPTIONAL IfcPositiveLengthMeasure;

```

WHERE

```

WR1 : FlangeThickness < (OverallDepth / 2.);
WR2 : WebThickness < OverallWidth;
WR3 : NOT(EXISTS(FilletRadius)) OR
      ((FilletRadius <= (OverallWidth - WebThickness)/2.) AND
       (FilletRadius <= (OverallDepth - (2. * FlangeThickness))/2.));

```

END_ENTITY;

ENTITY IfcJunctionBoxType

SUBTYPE OF (IfcFlowFittingType);

PredefinedType : IfcJunctionBoxTypeEnum;

END_ENTITY;

ENTITY IfcLShapeProfileDef

SUBTYPE OF (IfcParameterizedProfileDef);

```

Depth : IfcPositiveLengthMeasure;
Width : OPTIONAL IfcPositiveLengthMeasure;
Thickness : IfcPositiveLengthMeasure;
FilletRadius : OPTIONAL IfcPositiveLengthMeasure;
EdgeRadius : OPTIONAL IfcPositiveLengthMeasure;
LegSlope : OPTIONAL IfcPlaneAngleMeasure;
CentreOfGravityInX : OPTIONAL IfcPositiveLengthMeasure;
CentreOfGravityInY : OPTIONAL IfcPositiveLengthMeasure;

```

WHERE

```

WR21 : Thickness < Depth;
WR22 : NOT(EXISTS(Width)) OR (Thickness < Width);

```

END_ENTITY;

ENTITY IfcLampType

SUBTYPE OF (IfcFlowTerminalType);

PredefinedType : IfcLampTypeEnum;

END_ENTITY;

ENTITY IfcLightFixtureType

SUBTYPE OF (IfcFlowTerminalType);

PredefinedType : IfcLightFixtureTypeEnum;

END_ENTITY;

ENTITY IfcLine

SUBTYPE OF (IfcCurve);

Pnt : IfcCartesianPoint;

Dir : IfcVector;

WHERE

WR1 : Dir.Dim = Pnt.Dim;

END_ENTITY;

ENTITY IfcLocalPlacement

SUBTYPE OF (IfcObjectPlacement);

PlacementRelTo : OPTIONAL IfcObjectPlacement;

RelativePlacement : IfcAxis2Placement;

WHERE

WR21 : IfcCorrectLocalPlacement(RelativePlacement, PlacementRelTo);

END_ENTITY;

```
ENTITY IfcLoop
  SUPERTYPE OF (ONEOF
    (IfcPolyLoop))
  SUBTYPE OF (IfcTopologicalRepresentationItem);
END_ENTITY;
```

```
ENTITY IfcManifoldSolidBrep
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcFacetedBrep))
  SUBTYPE OF (IfcSolidModel);
  Outer : IfcClosedShell;
END_ENTITY;
```

```
ENTITY IfcMappedItem
  SUBTYPE OF (IfcRepresentationItem);
  MappingSource : IfcRepresentationMap;
  MappingTarget : IfcCartesianTransformationOperator;
END_ENTITY;
```

```
ENTITY IfcMaterial;
  Name : IfcLabel;
  INVERSE
    HasRepresentation : SET [0:1] OF IfcMaterialDefinitionRepresentation FOR
    RepresentedMaterial;
    ClassifiedAs : SET [0:1] OF IfcMaterialClassificationRelationship FOR ClassifiedMaterial;
END_ENTITY;
```

```
ENTITY IfcMaterialClassificationRelationship;
  MaterialClassifications : SET [1:?] OF IfcClassificationNotationSelect;
  ClassifiedMaterial : IfcMaterial;
END_ENTITY;
```

```
ENTITY IfcMaterialDefinitionRepresentation
  SUBTYPE OF (IfcProductRepresentation);
  RepresentedMaterial : IfcMaterial;
  WHERE
    WR11 : SIZEOF(QUERY(temp <* Representations |
      (NOT('IFC2X3_CV_V2_0.IFCSTYLEDREPRESENTATION' IN TYPEOF(temp)))
    )) = 0;
END_ENTITY;
```

```
ENTITY IfcMaterialLayer;
  Material : OPTIONAL IfcMaterial;
  LayerThickness : IfcPositiveLengthMeasure;
  IsVentilated : OPTIONAL IfcLogical;
  INVERSE
    ToMaterialLayerSet : IfcMaterialLayerSet FOR MaterialLayers;
END_ENTITY;
```

```
ENTITY IfcMaterialLayerSet;
  MaterialLayers : LIST [1:?] OF IfcMaterialLayer;
  LayerSetName : OPTIONAL IfcLabel;
  DERIVE
    TotalThickness : IfcLengthMeasure := IfcMlsTotalThickness(SELF);
END_ENTITY;
```

```

ENTITY IfcMaterialLayerSetUsage;
  ForLayerSet : IfcMaterialLayerSet;
  LayerSetDirection : IfcLayerSetDirectionEnum;
  DirectionSense : IfcDirectionSenseEnum;
  OffsetFromReferenceLine : IfcLengthMeasure;
END_ENTITY;

```

```

ENTITY IfcMaterialList;
  Materials : LIST [1:?] OF IfcMaterial;
END_ENTITY;

```

```

ENTITY IfcMeasureWithUnit;
  ValueComponent : IfcValue;
  UnitComponent : IfcUnit;
END_ENTITY;

```

```

ENTITY IfcMechanicalFastener
  SUBTYPE OF (IfcFastener);
  NominalDiameter : OPTIONAL IfcPositiveLengthMeasure;
  NominalLength : OPTIONAL IfcPositiveLengthMeasure;
END_ENTITY;

```

```

ENTITY IfcMechanicalFastenerType
  SUBTYPE OF (IfcFastenerType);
END_ENTITY;

```

```

ENTITY IfcMember
  SUBTYPE OF (IfcBuildingElement);
END_ENTITY;

```

```

ENTITY IfcMemberType
  SUBTYPE OF (IfcBuildingElementType);
  PredefinedType : IfcMemberTypeEnum;
END_ENTITY;

```

```

ENTITY IfcMotorConnectionType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
  PredefinedType : IfcMotorConnectionTypeEnum;
END_ENTITY;

```

```

ENTITY IfcNamedUnit
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcConversionBasedUnit
    ,IfcSIUnit));
  Dimensions : IfcDimensionalExponents;
  UnitType : IfcUnitEnum;
  WHERE
    WR1 : IfcCorrectDimensions (SELF.UnitType, SELF.Dimensions);
END_ENTITY;

```

```

ENTITY IfcObject
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcGroup
    ,IfcProduct
    ,IfcProject))
  SUBTYPE OF (IfcObjectDefinition);

```

```

    ObjectType : OPTIONAL IfcLabel;
INVERSE
    IsDefinedBy : SET [0:?] OF IfcRelDefines FOR RelatedObjects;
WHERE
    WR1 : SIZEOF(QUERY(temp <* IsDefinedBy | 'IFC2X3_CV_V2_0.IFCRELDEFINESBYTYPE' IN
        TYPEOF(temp))) <= 1;
END_ENTITY;

```

```

ENTITY IfcObjectDefinition
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcObject
        ,IfcTypeObject))
    SUBTYPE OF (IfcRoot);
INVERSE
    HasAssignments : SET [0:?] OF IfcRelAssigns FOR RelatedObjects;
    IsDecomposedBy : SET [0:?] OF IfcRelDecomposes FOR RelatingObject;
    Decomposes : SET [0:1] OF IfcRelDecomposes FOR RelatedObjects;
    HasAssociations : SET [0:?] OF IfcRelAssociates FOR RelatedObjects;
END_ENTITY;

```

```

ENTITY IfcObjectPlacement
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcLocalPlacement));
INVERSE
    PlacesObject : SET [1:1] OF IfcProduct FOR ObjectPlacement;
    ReferencedByPlacements : SET [0:?] OF IfcLocalPlacement FOR PlacementRelTo;
END_ENTITY;

```

```

ENTITY IfcOpenShell
    SUBTYPE OF (IfcConnectedFaceSet);
END_ENTITY;

```

```

ENTITY IfcOpeningElement
    SUBTYPE OF (IfcFeatureElementSubtraction);
INVERSE
    HasFillings : SET [0:?] OF IfcRelFillsElement FOR RelatingOpeningElement;
END_ENTITY;

```

```

ENTITY IfcOrganization;
    Id : OPTIONAL IfcIdentifier;
    Name : IfcLabel;
    Description : OPTIONAL IfcText;
    Roles : OPTIONAL LIST [1:?] OF IfcActorRole;
    Addresses : OPTIONAL LIST [1:?] OF IfcAddress;
INVERSE
    Engages : SET [0:?] OF IfcPersonAndOrganization FOR TheOrganization;
END_ENTITY;

```

```

ENTITY IfcOutletType
    SUBTYPE OF (IfcFlowTerminalType);
    PredefinedType : IfcOutletTypeEnum;
END_ENTITY;

```

```

ENTITY IfcOwnerHistory;
    OwningUser : IfcPersonAndOrganization;
    OwningApplication : IfcApplication;
    State : OPTIONAL IfcStateEnum;

```

```

ChangeAction : IfcChangeActionEnum;
LastModifiedDate : OPTIONAL IfcTimeStamp;
LastModifyingUser : OPTIONAL IfcPersonAndOrganization;
LastModifyingApplication : OPTIONAL IfcApplication;
CreationDate : IfcTimeStamp;

```

END_ENTITY;

ENTITY IfcParameterizedProfileDef

```

ABSTRACT SUPERTYPE OF (ONEOF
  (IfcCShapeProfileDef
  ,IfcCircleProfileDef
  ,IfcEllipseProfileDef
  ,IfcIShapeProfileDef
  ,IfcLShapeProfileDef
  ,IfcRectangleProfileDef
  ,IfcTShapeProfileDef
  ,IfcUShapeProfileDef
  ,IfcZShapeProfileDef))
SUBTYPE OF (IfcProfileDef);
  Position : IfcAxis2Placement2D;

```

END_ENTITY;

ENTITY IfcPerson;

```

  Id : OPTIONAL IfcIdentifier;
  FamilyName : OPTIONAL IfcLabel;
  GivenName : OPTIONAL IfcLabel;
  MiddleNames : OPTIONAL LIST [1:?] OF IfcLabel;
  PrefixTitles : OPTIONAL LIST [1:?] OF IfcLabel;
  SuffixTitles : OPTIONAL LIST [1:?] OF IfcLabel;
  Roles : OPTIONAL LIST [1:?] OF IfcActorRole;
  Addresses : OPTIONAL LIST [1:?] OF IfcAddress;

```

INVERSE

```

  EngagedIn : SET [0:?] OF IfcPersonAndOrganization FOR ThePerson;

```

WHERE

```

  WR1 : EXISTS(FamilyName) OR
        EXISTS(GivenName);

```

END_ENTITY;

ENTITY IfcPersonAndOrganization;

```

  ThePerson : IfcPerson;
  TheOrganization : IfcOrganization;
  Roles : OPTIONAL LIST [1:?] OF IfcActorRole;

```

END_ENTITY;

ENTITY IfcPile

```

SUBTYPE OF (IfcBuildingElement);
  PredefinedType : IfcPileTypeEnum;
  ConstructionType : OPTIONAL IfcPileConstructionEnum;
WHERE
  WR1 : (PredefinedType <> IfcPileTypeEnum.USERDEFINED) OR ((PredefinedType =
    IfcPileTypeEnum.USERDEFINED) AND EXISTS(SELF\IfcObject.ObjectType));

```

END_ENTITY;

ENTITY IfcPipeFittingType

```

SUBTYPE OF (IfcFlowFittingType);
  PredefinedType : IfcPipeFittingTypeEnum;

```

WHERE

```

    WR1 : (PredefinedType <> IfcPipeFittingTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcPipeFittingTypeEnum.USERDEFINED) AND
           EXISTS(SELf\IfcElementType.ElementType));

```

```

END_ENTITY;

```

```

ENTITY IfcPipeSegmentType

```

```

  SUBTYPE OF (IfcFlowSegmentType);

```

```

    PredefinedType : IfcPipeSegmentTypeEnum;

```

```

  WHERE

```

```

    WR1 : (PredefinedType <> IfcPipeSegmentTypeEnum.USERDEFINED) OR
          ((PredefinedType = IfcPipeSegmentTypeEnum.USERDEFINED) AND
           EXISTS(SELf\IfcElementType.ElementType));

```

```

END_ENTITY;

```

```

ENTITY IfcPlacement

```

```

  ABSTRACT SUPERTYPE OF (ONEOF

```

```

    (IfcAxis1Placement
     ,IfcAxis2Placement2D
     ,IfcAxis2Placement3D))

```

```

  SUBTYPE OF (IfcGeometricRepresentationItem);

```

```

    Location : IfcCartesianPoint;

```

```

  DERIVE

```

```

    Dim : IfcDimensionCount := Location.Dim;

```

```

END_ENTITY;

```

```

ENTITY IfcPlane

```

```

  SUBTYPE OF (IfcElementarySurface);

```

```

END_ENTITY;

```

```

ENTITY IfcPlate

```

```

  SUBTYPE OF (IfcBuildingElement);

```

```

END_ENTITY;

```

```

ENTITY IfcPlateType

```

```

  SUBTYPE OF (IfcBuildingElementType);

```

```

    PredefinedType : IfcPlateTypeEnum;

```

```

END_ENTITY;

```

```

ENTITY IfcPoint

```

```

  ABSTRACT SUPERTYPE OF (ONEOF

```

```

    (IfcCartesianPoint))

```

```

  SUBTYPE OF (IfcGeometricRepresentationItem);

```

```

END_ENTITY;

```

```

ENTITY IfcPolyLoop

```

```

  SUBTYPE OF (IfcLoop);

```

```

    Polygon : LIST [3:?] OF UNIQUE IfcCartesianPoint;

```

```

  WHERE

```

```

    WR21 : SIZEOF(QUERY(Temp <* Polygon | Temp.Dim <> Polygon[1].Dim)) = 0;

```

```

END_ENTITY;

```

```

ENTITY IfcPolygonalBoundedHalfSpace

```

```

  SUBTYPE OF (IfcHalfSpaceSolid);

```

```

    Position : IfcAxis2Placement3D;

```

```

    PolygonalBoundary : IfcBoundedCurve;

```

```

  WHERE

```

```

    WR41 : PolygonalBoundary.Dim = 2;

```

```

WR42 : SIZEOF(TYPEOF(PolygonalBoundary) * [
    'IFC2X3_CV_V2_0.IFCPOLYLINE',
    'IFC2X3_CV_V2_0.IFCCOMPOSITECURVE'])
) = 1;

```

```
END_ENTITY;
```

```
ENTITY IfcPolyline
```

```
  SUBTYPE OF (IfcBoundedCurve);
```

```
  Points : LIST [2:?] OF IfcCartesianPoint;
```

```
  WHERE
```

```
    WR41 : SIZEOF(QUERY(Temp <* Points | Temp.Dim <> Points[1].Dim)) = 0;
```

```
END_ENTITY;
```

```
ENTITY IfcPort
```

```
  ABSTRACT SUPERTYPE OF (ONEOF
```

```
    (IfcDistributionPort))
```

```
  SUBTYPE OF (IfcProduct);
```

```
  INVERSE
```

```
    ContainedIn : IfcRelConnectsPortToElement FOR RelatingPort;
```

```
    ConnectedFrom : SET [0:1] OF IfcRelConnectsPorts FOR RelatedPort;
```

```
    ConnectedTo : SET [0:1] OF IfcRelConnectsPorts FOR RelatingPort;
```

```
END_ENTITY;
```

```
ENTITY IfcPostalAddress
```

```
  SUBTYPE OF (IfcAddress);
```

```
    InternalLocation : OPTIONAL IfcLabel;
```

```
    AddressLines : OPTIONAL LIST [1:?] OF IfcLabel;
```

```
    PostalBox : OPTIONAL IfcLabel;
```

```
    Town : OPTIONAL IfcLabel;
```

```
    Region : OPTIONAL IfcLabel;
```

```
    PostalCode : OPTIONAL IfcLabel;
```

```
    Country : OPTIONAL IfcLabel;
```

```
  WHERE
```

```
    WR1 : EXISTS (InternalLocation) OR
```

```
        EXISTS (AddressLines) OR
```

```
        EXISTS (PostalBox) OR
```

```
        EXISTS (PostalCode) OR
```

```
        EXISTS (Town) OR
```

```
        EXISTS (Region) OR
```

```
        EXISTS (Country);
```

```
END_ENTITY;
```

```
ENTITY IfcPresentationLayerAssignment
```

```
  SUPERTYPE OF (ONEOF
```

```
    (IfcPresentationLayerWithStyle));
```

```
  Name : IfcLabel;
```

```
  Description : OPTIONAL IfcText;
```

```
  AssignedItems : SET [1:?] OF IfcLayeredItem;
```

```
  Identifier : OPTIONAL IfcIdentifier;
```

```
END_ENTITY;
```

```
ENTITY IfcPresentationLayerWithStyle
```

```
  SUBTYPE OF (IfcPresentationLayerAssignment);
```

```
    LayerOn : LOGICAL;
```

```
    LayerFrozen : LOGICAL;
```

```
    LayerBlocked : LOGICAL;
```

```
    LayerStyles : SET [0:?] OF IfcPresentationStyleSelect;
```

END_ENTITY;

```
ENTITY IfcPresentationStyle
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcCurveStyle
    ,IfcSurfaceStyle));
  Name : OPTIONAL IfcLabel;
END_ENTITY;
```

```
ENTITY IfcPresentationStyleAssignment;
  Styles : SET [1:?] OF IfcPresentationStyleSelect;
END_ENTITY;
```

```
ENTITY IfcProduct
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcElement
    ,IfcGrid
    ,IfcPort
    ,IfcSpatialStructureElement))
  SUBTYPE OF (IfcObject);
  ObjectPlacement : OPTIONAL IfcObjectPlacement;
  Representation : OPTIONAL IfcProductRepresentation;
  WHERE
    WR1 : (EXISTS(Representation) AND EXISTS(ObjectPlacement))
          OR (EXISTS(Representation) AND
              (NOT('IFC2X3_CV_V2_0.IFCPRODUCTDEFINITIONSHAPE' IN
                  TYPEOF(Representation))))
          OR (NOT(EXISTS(Representation)));
END_ENTITY;
```

```
ENTITY IfcProductDefinitionShape
  SUBTYPE OF (IfcProductRepresentation);
  INVERSE
    ShapeOfProduct : SET [1:1] OF IfcProduct FOR Representation;
  WHERE
    WR11 : SIZEOF(QUERY(temp <* Representations |
      (NOT('IFC2X3_CV_V2_0.IFCSHAPEMODEL' IN TYPEOF(temp)))
    )) = 0;
END_ENTITY;
```

```
ENTITY IfcProductRepresentation
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcMaterialDefinitionRepresentation
    ,IfcProductDefinitionShape));
  Name : OPTIONAL IfcLabel;
  Description : OPTIONAL IfcText;
  Representations : LIST [1:?] OF IfcRepresentation;
END_ENTITY;
```

```
ENTITY IfcProfileDef
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcArbitraryClosedProfileDef
    ,IfcArbitraryOpenProfileDef
    ,IfcCompositeProfileDef
    ,IfcDerivedProfileDef
    ,IfcParameterizedProfileDef));
  ProfileType : IfcProfileTypeEnum;
```

```

    ProfileName : OPTIONAL IfcLabel;
END_ENTITY;

ENTITY IfcProject
  SUBTYPE OF (IfcObject);
    LongName : OPTIONAL IfcLabel;
    Phase : OPTIONAL IfcLabel;
    RepresentationContexts : SET [1:?] OF IfcRepresentationContext;
    UnitsInContext : IfcUnitAssignment;
  WHERE
    WR31 : EXISTS(SELF\IfcRoot.Name);
    WR32 : SIZEOF(QUERY(Temp <* RepresentationContexts |
      'IFC2X3_CV_V2_0.IFCGEOMETRICREPRESENTATIONSUBCONTEXT' IN TYPEOF(Temp)
    )) = 0;
    WR33 : SIZEOF(SELF\IfcObjectDefinition.Decomposes) = 0;
END_ENTITY;

ENTITY IfcProperty
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcComplexProperty
    ,IfcSimpleProperty));
    Name : IfcIdentifier;
    Description : OPTIONAL IfcText;
  INVERSE
    PartOfComplex : SET [0:1] OF IfcComplexProperty FOR HasProperties;
END_ENTITY;

ENTITY IfcPropertyBoundedValue
  SUBTYPE OF (IfcSimpleProperty);
    UpperBoundValue : OPTIONAL IfcValue;
    LowerBoundValue : OPTIONAL IfcValue;
    Unit : OPTIONAL IfcUnit;
  WHERE
    WR21 : NOT(EXISTS(UpperBoundValue)) OR NOT(EXISTS(LowerBoundValue)) OR
      (TYPEOF(UpperBoundValue) = TYPEOF(LowerBoundValue));
    WR22 : EXISTS(UpperBoundValue) OR EXISTS(LowerBoundValue);
END_ENTITY;

ENTITY IfcPropertyDefinition
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcPropertySetDefinition))
  SUBTYPE OF (IfcRoot);
  INVERSE
    HasAssociations : SET [0:?] OF IfcRelAssociates FOR RelatedObjects;
END_ENTITY;

ENTITY IfcPropertyEnumeratedValue
  SUBTYPE OF (IfcSimpleProperty);
    EnumerationValues : LIST [1:?] OF IfcValue;
    EnumerationReference : OPTIONAL IfcPropertyEnumeration;
  WHERE
    WR1 : NOT(EXISTS(EnumerationReference)) OR
      (SIZEOF(QUERY(temp <* EnumerationValues |
        temp IN EnumerationReference.EnumerationValues))
      = SIZEOF(EnumerationValues));
END_ENTITY;

```

```

ENTITY IfcPropertyEnumeration;
    Name : IfcLabel;
    EnumerationValues : LIST [1:?] OF UNIQUE IfcValue;
    Unit : OPTIONAL IfcUnit;
UNIQUE
    UR1 : Name;
WHERE
    WR01 : SIZEOF(QUERY(temp <* SELF.EnumerationValues |
        NOT(TYPEOF(SELF.EnumerationValues[1]) = TYPEOF(temp))
    )) = 0;
END_ENTITY;

```

```

ENTITY IfcPropertyListValue
    SUBTYPE OF (IfcSimpleProperty);
    ListValues : LIST [1:?] OF IfcValue;
    Unit : OPTIONAL IfcUnit;
WHERE
    WR31 : SIZEOF(QUERY(temp <* SELF.ListValues |
        NOT(TYPEOF(SELF.ListValues[1]) = TYPEOF(temp))
    )) = 0;
END_ENTITY;

```

```

ENTITY IfcPropertySet
    SUBTYPE OF (IfcPropertySetDefinition);
    HasProperties : SET [1:?] OF IfcProperty;
WHERE
    WR31 : EXISTS(SELF\IfcRoot.Name);
    WR32 : IfcUniquePropertyName(HasProperties);
END_ENTITY;

```

```

ENTITY IfcPropertySetDefinition
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcDoorLiningProperties
        ,IfcDoorPanelProperties
        ,IfcPropertySet
        ,IfcWindowLiningProperties
        ,IfcWindowPanelProperties))
    SUBTYPE OF (IfcPropertyDefinition);
    INVERSE
        PropertyDefinitionOf : SET [0:1] OF IfcRelDefinesByProperties FOR
        RelatingPropertyDefinition;
        DefinesType : SET [0:1] OF IfcTypeObject FOR HasPropertySets;
END_ENTITY;

```

```

ENTITY IfcPropertySingleValue
    SUBTYPE OF (IfcSimpleProperty);
    NominalValue : OPTIONAL IfcValue;
    Unit : OPTIONAL IfcUnit;
END_ENTITY;

```

```

ENTITY IfcPropertyTableValue
    SUBTYPE OF (IfcSimpleProperty);
    DefiningValues : LIST [1:?] OF UNIQUE IfcValue;
    DefinedValues : LIST [1:?] OF IfcValue;
    Expression : OPTIONAL IfcText;
    DefiningUnit : OPTIONAL IfcUnit;
    DefinedUnit : OPTIONAL IfcUnit;

```

```

WHERE
    WR1 : SIZEOF(DefiningValues) = SIZEOF(DefinedValues);
    WR2 : SIZEOF(QUERY(temp <* SELF.DefiningValues | TYPEOF(temp) <>
        TYPEOF(SELF.DefiningValues[1])))=0;
    WR3 : SIZEOF(QUERY(temp <* SELF.DefinedValues | TYPEOF(temp) <>
        TYPEOF(SELF.DefinedValues[1])))=0;
END_ENTITY;

ENTITY IfcProtectiveDeviceType
    SUBTYPE OF (IfcFlowControllerType);
    PredefinedType : IfcProtectiveDeviceTypeEnum;
END_ENTITY;

ENTITY IfcPumpType
    SUBTYPE OF (IfcFlowMovingDeviceType);
    PredefinedType : IfcPumpTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcPumpTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcPumpTypeEnum.USERDEFINED) AND
        EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcRailing
    SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : OPTIONAL IfcRailingTypeEnum;
WHERE
    WR61 : NOT(EXISTS(PredefinedType)) OR
        (PredefinedType <> IfcRailingTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcRailingTypeEnum.USERDEFINED) AND EXISTS
        (SELF\IfcObject.ObjectType));
END_ENTITY;

ENTITY IfcRailingType
    SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : IfcRailingTypeEnum;
END_ENTITY;

ENTITY IfcRamp
    SUBTYPE OF (IfcBuildingElementType);
    ShapeType : IfcRampTypeEnum;
WHERE
    WR1 : (HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 0) OR
        ((HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 1) AND
        (NOT(EXISTS(SELF\IfcProduct.Representation))));
END_ENTITY;

ENTITY IfcRampFlight
    SUBTYPE OF (IfcBuildingElementType);
END_ENTITY;

ENTITY IfcRampFlightType
    SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : IfcRampFlightTypeEnum;
END_ENTITY;

ENTITY IfcRectangleHollowProfileDef
    SUBTYPE OF (IfcRectangleProfileDef);

```

```

WallThickness : IfcPositiveLengthMeasure;
InnerFilletRadius : OPTIONAL IfcPositiveLengthMeasure;
OuterFilletRadius : OPTIONAL IfcPositiveLengthMeasure;

```

WHERE

```

WR31 : (WallThickness < (SELF\IfcRectangleProfileDef.XDim/2.)) AND
        (WallThickness < (SELF\IfcRectangleProfileDef.YDim/2.));
WR32 : NOT(EXISTS(OuterFilletRadius)) OR
        ((OuterFilletRadius <= (SELF\IfcRectangleProfileDef.XDim/2.)) AND
         (OuterFilletRadius <= (SELF\IfcRectangleProfileDef.YDim/2.)));
WR33 : NOT(EXISTS(InnerFilletRadius)) OR
        ((InnerFilletRadius <= (SELF\IfcRectangleProfileDef.XDim/2. - WallThickness))
         AND
         (InnerFilletRadius <= (SELF\IfcRectangleProfileDef.YDim/2. - WallThickness)));

```

END_ENTITY;

ENTITY IfcRectangleProfileDef

```

SUPERTYPE OF (ONEOF
  (IfcRectangleHollowProfileDef
   ,IfcRoundedRectangleProfileDef))
SUBTYPE OF (IfcParameterizedProfileDef);
XDim : IfcPositiveLengthMeasure;
YDim : IfcPositiveLengthMeasure;

```

END_ENTITY;

ENTITY IfcReinforcingBar

```

SUBTYPE OF (IfcReinforcingElement);
NominalDiameter : IfcPositiveLengthMeasure;
CrossSectionArea : IfcAreaMeasure;
BarLength : OPTIONAL IfcPositiveLengthMeasure;
BarRole : IfcReinforcingBarRoleEnum;
BarSurface : OPTIONAL IfcReinforcingBarSurfaceEnum;

```

WHERE

```

WR1 : (BarRole <> IfcReinforcingBarRoleEnum.USERDEFINED) OR ((BarRole =
  IfcReinforcingBarRoleEnum.USERDEFINED) AND EXISTS(SELF\IfcObject.ObjectType));

```

END_ENTITY;

ENTITY IfcReinforcingElement

```

ABSTRACT SUPERTYPE OF (ONEOF
  (IfcReinforcingBar
   ,IfcReinforcingMesh
   ,IfcTendon
   ,IfcTendonAnchor))
SUBTYPE OF (IfcBuildingElementComponent);
SteelGrade : OPTIONAL IfcLabel;

```

END_ENTITY;

ENTITY IfcReinforcingMesh

```

SUBTYPE OF (IfcReinforcingElement);
MeshLength : OPTIONAL IfcPositiveLengthMeasure;
MeshWidth : OPTIONAL IfcPositiveLengthMeasure;
LongitudinalBarNominalDiameter : IfcPositiveLengthMeasure;
TransverseBarNominalDiameter : IfcPositiveLengthMeasure;
LongitudinalBarCrossSectionArea : IfcAreaMeasure;
TransverseBarCrossSectionArea : IfcAreaMeasure;
LongitudinalBarSpacing : IfcPositiveLengthMeasure;
TransverseBarSpacing : IfcPositiveLengthMeasure;

```

END_ENTITY;

```
ENTITY IfcRelAggregates
  SUBTYPE OF (IfcRelDecomposes);
END_ENTITY;
```

```
ENTITY IfcRelAssigns
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcRelAssignsToGroup))
  SUBTYPE OF (IfcRelationship);
    RelatedObjects : SET [1:?] OF IfcObjectDefinition;
    RelatedObjectsType : OPTIONAL IfcObjectTypeEnum;
  WHERE
    WR1 : IfcCorrectObjectAssignment(RelatedObjectsType, RelatedObjects);
END_ENTITY;
```

```
ENTITY IfcRelAssignsToGroup
  SUBTYPE OF (IfcRelAssigns);
    RelatingGroup : IfcGroup;
  WHERE
    WR1 : SIZEOF(QUERY(Temp <* SELF\IfcRelAssigns.RelatedObjects | RelatingGroup == Temp))
    = 0;
END_ENTITY;
```

```
ENTITY IfcRelAssociates
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcRelAssociatesClassification
    ,IfcRelAssociatesMaterial))
  SUBTYPE OF (IfcRelationship);
    RelatedObjects : SET [1:?] OF IfcRoot;
  WHERE
    WR21 : SIZEOF(QUERY(temp <* RelatedObjects |
      NOT(('IFC2X3_CV_V2_0.IFCOBJECTDEFINITION' IN TYPEOF(temp))
      OR ('IFC2X3_CV_V2_0.IFCPROPERTYDEFINITION' IN TYPEOF(temp)))
    )) = 0;
END_ENTITY;
```

```
ENTITY IfcRelAssociatesClassification
  SUBTYPE OF (IfcRelAssociates);
    RelatingClassification : IfcClassificationNotationSelect;
END_ENTITY;
```

```
ENTITY IfcRelAssociatesMaterial
  SUBTYPE OF (IfcRelAssociates);
    RelatingMaterial : IfcMaterialSelect;
  WHERE
    WR21 : SIZEOF(QUERY(temp <* SELF\IfcRelAssociates.RelatedObjects |
      ('IFC2X3_CV_V2_0.IFCFEATUREELEMENTSUBTRACTION' IN TYPEOF(temp))
      OR ('IFC2X3_CV_V2_0.IFCVIRTUALELEMENT' IN TYPEOF(temp))
    )) = 0;
    WR22 : SIZEOF(QUERY(temp <* SELF\IfcRelAssociates.RelatedObjects |
      (NOT('IFC2X3_CV_V2_0.IFCPRODUCT' IN TYPEOF(temp))
      AND NOT('IFC2X3_CV_V2_0.IFCTYPEPRODUCT' IN TYPEOF(temp)))
    )) = 0;
END_ENTITY;
```

```
ENTITY IfcRelConnects
  ABSTRACT SUPERTYPE OF (ONEOF
```

```

    (IfcRelConnectsElements
    ,IfcRelConnectsPortToElement
    ,IfcRelConnectsPorts
    ,IfcRelContainedInSpatialStructure
    ,IfcRelFillsElement
    ,IfcRelFlowControlElements
    ,IfcRelServicesBuildings
    ,IfcRelVoidsElement))
SUBTYPE OF (IfcRelationship);
END_ENTITY;

ENTITY IfcRelConnectsElements
SUPERTYPE OF (ONEOF
    (IfcRelConnectsPathElements
    ,IfcRelConnectsWithRealizingElements))
SUBTYPE OF (IfcRelConnects);
    ConnectionGeometry : OPTIONAL IfcConnectionGeometry;
    RelatingElement : IfcElement;
    RelatedElement : IfcElement;
WHERE
    WR31 : RelatingElement :<>: RelatedElement;
END_ENTITY;

ENTITY IfcRelConnectsPathElements
SUBTYPE OF (IfcRelConnectsElements);
    RelatingPriorities : LIST [0:?] OF INTEGER;
    RelatedPriorities : LIST [0:?] OF INTEGER;
    RelatedConnectionType : IfcConnectionTypeEnum;
    RelatingConnectionType : IfcConnectionTypeEnum;
END_ENTITY;

ENTITY IfcRelConnectsPortToElement
SUBTYPE OF (IfcRelConnects);
    RelatingPort : IfcPort;
    RelatedElement : IfcElement;
END_ENTITY;

ENTITY IfcRelConnectsPorts
SUBTYPE OF (IfcRelConnects);
    RelatingPort : IfcPort;
    RelatedPort : IfcPort;
    RealizingElement : OPTIONAL IfcElement;
END_ENTITY;

ENTITY IfcRelConnectsWithRealizingElements
SUBTYPE OF (IfcRelConnectsElements);
    RealizingElements : SET [1:?] OF IfcElement;
    ConnectionType : OPTIONAL IfcLabel;
END_ENTITY;

ENTITY IfcRelContainedInSpatialStructure
SUBTYPE OF (IfcRelConnects);
    RelatedElements : SET [1:?] OF IfcProduct;
    RelatingStructure : IfcSpatialStructureElement;
WHERE
    WR31 : SIZEOF(QUERY(temp <* RelatedElements |
    'IFC2X3_CV_V2_0.IFCSPATIALSTRUCTUREELEMENT' IN TYPEOF(temp))) = 0;

```

END_ENTITY;

```
ENTITY IfcRelCoversBldgElements
  SUBTYPE OF (IfcRelConnects);
    RelatingBuildingElement : IfcElement;
    RelatedCoverings : SET [1:?] OF IfcCovering;
END_ENTITY;
```

```
ENTITY IfcRelDecomposes
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcRelAggregates))
  SUBTYPE OF (IfcRelationship);
    RelatingObject : IfcObjectDefinition;
    RelatedObjects : SET [1:?] OF IfcObjectDefinition;
  WHERE
    WR31 : SIZEOF(QUERY(Temp <* RelatedObjects | RelatingObject := Temp)) = 0;
END_ENTITY;
```

```
ENTITY IfcRelDefines
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcRelDefinesByProperties
    ,IfcRelDefinesByType))
  SUBTYPE OF (IfcRelationship);
    RelatedObjects : SET [1:?] OF IfcObject;
END_ENTITY;
```

```
ENTITY IfcRelDefinesByProperties
  SUBTYPE OF (IfcRelDefines);
    RelatingPropertyDefinition : IfcPropertySetDefinition;
END_ENTITY;
```

```
ENTITY IfcRelDefinesByType
  SUBTYPE OF (IfcRelDefines);
    RelatingType : IfcTypeObject;
END_ENTITY;
```

```
ENTITY IfcRelFillsElement
  SUBTYPE OF (IfcRelConnects);
    RelatingOpeningElement : IfcOpeningElement;
    RelatedBuildingElement : IfcElement;
END_ENTITY;
```

```
ENTITY IfcRelFlowControlElements
  SUBTYPE OF (IfcRelConnects);
    RelatedControlElements : SET [1:?] OF IfcDistributionControlElement;
    RelatingFlowElement : IfcDistributionFlowElement;
END_ENTITY;
```

```
ENTITY IfcRelServicesBuildings
  SUBTYPE OF (IfcRelConnects);
    RelatingSystem : IfcSystem;
    RelatedBuildings : SET [1:?] OF IfcSpatialStructureElement;
END_ENTITY;
```

```
ENTITY IfcRelVoidsElement
  SUBTYPE OF (IfcRelConnects);
    RelatingBuildingElement : IfcElement;
```

```

    RelatedOpeningElement : IfcFeatureElementSubtraction;
END_ENTITY;

```

```

ENTITY IfcRelationship
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcRelAssigns
    ,IfcRelAssociates
    ,IfcRelConnects
    ,IfcRelDecomposes
    ,IfcRelDefines))
  SUBTYPE OF (IfcRoot);
END_ENTITY;

```

```

ENTITY IfcRepresentation
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcShapeModel
    ,IfcStyleModel));
  ContextOfItems : IfcRepresentationContext;
  RepresentationIdentifier : OPTIONAL IfcLabel;
  RepresentationType : OPTIONAL IfcLabel;
  Items : SET [1:?] OF IfcRepresentationItem;
  INVERSE
    RepresentationMap : SET [0:1] OF IfcRepresentationMap FOR MappedRepresentation;
    LayerAssignments : SET [0:?] OF IfcPresentationLayerAssignment FOR AssignedItems;
    OfProductRepresentation : SET [0:1] OF IfcProductRepresentation FOR Representations;
END_ENTITY;

```

```

ENTITY IfcRepresentationContext
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcGeometricRepresentationContext));
  ContextIdentifier : OPTIONAL IfcLabel;
  ContextType : OPTIONAL IfcLabel;
  INVERSE
    RepresentationsInContext : SET [0:?] OF IfcRepresentation FOR ContextOfItems;
END_ENTITY;

```

```

ENTITY IfcRepresentationItem
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcGeometricRepresentationItem
    ,IfcMappedItem
    ,IfcStyledItem
    ,IfcTopologicalRepresentationItem));
  INVERSE
    LayerAssignments : SET [0:?] OF IfcPresentationLayerAssignment FOR AssignedItems;
    StyledByItem : SET [0:1] OF IfcStyledItem FOR Item;
END_ENTITY;

```

```

ENTITY IfcRepresentationMap;
  MappingOrigin : IfcAxis2Placement;
  MappedRepresentation : IfcRepresentation;
  INVERSE
    MapUsage : SET [0:?] OF IfcMappedItem FOR MappingSource;
END_ENTITY;

```

```

ENTITY IfcRevolvedAreaSolid
  SUBTYPE OF (IfcSweptAreaSolid);
  Axis : IfcAxis1Placement;

```

```

    Angle : IfcPlaneAngleMeasure;
DERIVE
    AxisLine : IfcLine := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
    IfcCurve() || IfcLine(Axis.Location,
        IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcVector(Axis.Z,1.0));
WHERE
    WR31 : Axis.Location.Coordinates[3] = 0.0;
    WR32 : Axis.Z.DirectionRatios[3] = 0.0;
END_ENTITY;

```

```

ENTITY IfcRoof
    SUBTYPE OF (IfcBuildingElement);
    ShapeType : IfcRoofTypeEnum;
WHERE
    WR1 : (HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 0) OR
        ((HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 1) AND
        (NOT(EXISTS(SELF\IfcProduct.Representation))));
END_ENTITY;

```

```

ENTITY IfcRoot
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcObjectDefinition
        ,IfcPropertyDefinition
        ,IfcRelationship));
    GlobalId : IfcGloballyUniqueId;
    OwnerHistory : IfcOwnerHistory;
    Name : OPTIONAL IfcLabel;
    Description : OPTIONAL IfcText;
    UNIQUE
        UR1 : GlobalId;
END_ENTITY;

```

```

ENTITY IfcRoundedRectangleProfileDef
    SUBTYPE OF (IfcRectangleProfileDef);
    RoundingRadius : IfcPositiveLengthMeasure;
WHERE
    WR31 : ((RoundingRadius <= (SELF\IfcRectangleProfileDef.XDim/2.)) AND
        (RoundingRadius <= (SELF\IfcRectangleProfileDef.YDim/2.)));
END_ENTITY;

```

```

ENTITY IfcSIUnit
    SUBTYPE OF (IfcNamedUnit);
    Prefix : OPTIONAL IfcSIPrefix;
    Name : IfcSIUnitName;
DERIVE
    SELF\IfcNamedUnit.Dimensions : IfcDimensionalExponents := IfcDimensionsForSiUnit
        (SELF.Name);
END_ENTITY;

```

```

ENTITY IfcSanitaryTerminalType
    SUBTYPE OF (IfcFlowTerminalType);
    PredefinedType : IfcSanitaryTerminalTypeEnum;
END_ENTITY;

```

```

ENTITY IfcSensorType
    SUBTYPE OF (IfcDistributionControlElementType);

```

```

    PredefinedType : IfcSensorTypeEnum;
END_ENTITY;

```

```

ENTITY IfcShapeModel
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcShapeRepresentation))
  SUBTYPE OF (IfcRepresentation);
  WHERE
    WR11 : (SIZEOF(SELF\IfcRepresentation.OfProductRepresentation) = 1) XOR
      (SIZEOF(SELF\IfcRepresentation.RepresentationMap) = 1);
END_ENTITY;

```

```

ENTITY IfcShapeRepresentation
  SUBTYPE OF (IfcShapeModel);
  WHERE
    WR21 : 'IFC2X3_CV_V2_0.IFCGEOMETRICREPRESENTATIONCONTEXT'
      IN TYPEOF(SELF\IfcRepresentation.ContextOfItems);
    WR22 : SIZEOF(QUERY(temp <* Items |
      ('IFC2X3_CV_V2_0.IFCTOPOLOGICALREPRESENTATIONITEM' IN TYPEOF(temp))
      AND (NOT(SIZEOF(
        ['IFC2X3_CV_V2_0.IFCVERTEXPOINT',
        'IFC2X3_CV_V2_0.IFCEDGECURVE',
        'IFC2X3_CV_V2_0.IFCFACESURFACE'] * TYPEOF(temp)) = 1))
      )) = 0;
    WR23 : EXISTS(SELF\IfcRepresentation.RepresentationType);
    WR24 : IfcShapeRepresentationTypes(SELF\IfcRepresentation.RepresentationType,
      SELF\IfcRepresentation.Items);
END_ENTITY;

```

```

ENTITY IfcShellBasedSurfaceModel
  SUBTYPE OF (IfcGeometricRepresentationItem);
  SbsmBoundary : SET [1:?] OF IfcShell;
  DERIVE
    Dim : IfcDimensionCount := 3;
END_ENTITY;

```

```

ENTITY IfcSimpleProperty
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcPropertyBoundedValue
    ,IfcPropertyEnumeratedValue
    ,IfcPropertyListValue
    ,IfcPropertySingleValue
    ,IfcPropertyTableValue))
  SUBTYPE OF (IfcProperty);
END_ENTITY;

```

```

ENTITY IfcSite
  SUBTYPE OF (IfcSpatialStructureElement);
  RefLatitude : OPTIONAL IfcCompoundPlaneAngleMeasure;
  RefLongitude : OPTIONAL IfcCompoundPlaneAngleMeasure;
  RefElevation : OPTIONAL IfcLengthMeasure;
  LandTitleNumber : OPTIONAL IfcLabel;
  SiteAddress : OPTIONAL IfcPostalAddress;
END_ENTITY;

```

```

ENTITY IfcSlab
  SUBTYPE OF (IfcBuildingElement);

```

```

    PredefinedType : OPTIONAL IfcSlabTypeEnum;
WHERE
    WR61 : NOT(EXISTS(PredefinedType)) OR
        (PredefinedType <> IfcSlabTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcSlabTypeEnum.USERDEFINED) AND EXISTS
            (SELF\IfcObject.ObjectType));
END_ENTITY;

ENTITY IfcSlabType
    SUBTYPE OF (IfcBuildingElementType);
    PredefinedType : IfcSlabTypeEnum;
END_ENTITY;

ENTITY IfcSolidModel
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcManifoldSolidBrep
        ,IfcSweptAreaSolid
        ,IfcSweptDiskSolid))
    SUBTYPE OF (IfcGeometricRepresentationItem);
    DERIVE
        Dim : IfcDimensionCount := 3;
END_ENTITY;

ENTITY IfcSpace
    SUBTYPE OF (IfcSpatialStructureElement);
    InteriorOrExteriorSpace : IfcInternalOrExternalEnum;
    ElevationWithFlooring : OPTIONAL IfcLengthMeasure;
END_ENTITY;

ENTITY IfcSpaceHeaterType
    SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcSpaceHeaterTypeEnum;
WHERE
    WR1 : (PredefinedType <> IfcSpaceHeaterTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcSpaceHeaterTypeEnum.USERDEFINED) AND
            EXISTS(SELF\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcSpaceType
    SUBTYPE OF (IfcSpatialStructureElementType);
    PredefinedType : IfcSpaceTypeEnum;
END_ENTITY;

ENTITY IfcSpatialStructureElement
    ABSTRACT SUPERTYPE OF (ONEOF
        (IfcBuilding
        ,IfcBuildingStorey
        ,IfcSite
        ,IfcSpace))
    SUBTYPE OF (IfcProduct);
    LongName : OPTIONAL IfcLabel;
    CompositionType : IfcElementCompositionEnum;
    INVERSE
        ServicedBySystems : SET [0:?] OF IfcRelServicesBuildings FOR RelatedBuildings;
        ContainsElements : SET [0:?] OF IfcRelContainedInSpatialStructure FOR RelatingStructure;
WHERE
    WR41 : (HIINDEX(SELF\IfcObjectDefinition.Decomposes) = 1)

```

```

AND
('IFC2X3_CV_V2_0.IFCRELAGGREGATES' IN
TYPEOF(SELF\IfcObjectDefinition.Decomposes[1]))
AND
(('IFC2X3_CV_V2_0.IFCPROJECT' IN TYPEOF
(SELF\IfcObjectDefinition.Decomposes[1].RelatingObject)) OR
('IFC2X3_CV_V2_0.IFCSPATIALSTRUCTUREELEMENT' IN TYPEOF
(SELF\IfcObjectDefinition.Decomposes[1].RelatingObject))
);

```

```
END_ENTITY;
```

```
ENTITY IfcSpatialStructureElementType
```

```
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcSpaceType))
```

```
  SUBTYPE OF (IfcElementType);
```

```
END_ENTITY;
```

```
ENTITY IfcStackTerminalType
```

```
  SUBTYPE OF (IfcFlowTerminalType);
```

```
    PredefinedType : IfcStackTerminalTypeEnum;
```

```
END_ENTITY;
```

```
ENTITY IfcStair
```

```
  SUBTYPE OF (IfcBuildingElement);
```

```
    ShapeType : IfcStairTypeEnum;
```

```
  WHERE
```

```
    WR1 : (HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 0) OR
      ((HIINDEX(SELF\IfcObjectDefinition.IsDecomposedBy) = 1) AND
      (NOT(EXISTS(SELF\IfcProduct.Representation))));
```

```
END_ENTITY;
```

```
ENTITY IfcStairFlight
```

```
  SUBTYPE OF (IfcBuildingElement);
```

```
    NumberOfRiser : OPTIONAL INTEGER;
```

```
    NumberOfTreads : OPTIONAL INTEGER;
```

```
    RiserHeight : OPTIONAL IfcPositiveLengthMeasure;
```

```
    TreadLength : OPTIONAL IfcPositiveLengthMeasure;
```

```
END_ENTITY;
```

```
ENTITY IfcStairFlightType
```

```
  SUBTYPE OF (IfcBuildingElementType);
```

```
    PredefinedType : IfcStairFlightTypeEnum;
```

```
END_ENTITY;
```

```
ENTITY IfcStyleModel
```

```
  ABSTRACT SUPERTYPE OF (ONEOF
```

```
    (IfcStyledRepresentation))
```

```
  SUBTYPE OF (IfcRepresentation);
```

```
END_ENTITY;
```

```
ENTITY IfcStyledItem
```

```
  SUBTYPE OF (IfcRepresentationItem);
```

```
    Item : OPTIONAL IfcRepresentationItem;
```

```
    Styles : SET [1:?] OF IfcPresentationStyleAssignment;
```

```
    Name : OPTIONAL IfcLabel;
```

```
  WHERE
```

```
    WR11 : SIZEOF(Styles) = 1;
```

```

WR12 : NOT('IFC2X3_CV_V2_0.IFCSTYLEDITEM' IN TYPEOF(Item));
END_ENTITY;

```

```

ENTITY IfcStyledRepresentation
  SUBTYPE OF (IfcStyleModel);
  WHERE
    WR21 : SIZEOF(QUERY(temp <* SELF\IfcRepresentation.Items |
      (NOT('IFC2X3_CV_V2_0.IFCSTYLEDITEM' IN TYPEOF(temp)))
    )) = 0;
END_ENTITY;

```

```

ENTITY IfcSurface
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcElementarySurface))
  SUBTYPE OF (IfcGeometricRepresentationItem);
END_ENTITY;

```

```

ENTITY IfcSurfaceStyle
  SUBTYPE OF (IfcPresentationStyle);
  Side : IfcSurfaceSide;
  Styles : SET [1:5] OF IfcSurfaceStyleElementSelect;
  WHERE
    WR11 : SIZEOF(QUERY(Style <* SELF.Styles |
      'IFC2X3_CV_V2_0.IFCSURFACESTYLESHADING' IN
      TYPEOF(Style)
    )) <= 1;
    WR12 : SIZEOF(QUERY(Style <* SELF.Styles |
      'IFC2X3_CV_V2_0.IFCSURFACESTYLELIGHTING' IN
      TYPEOF(Style)
    )) <= 1;
    WR13 : SIZEOF(QUERY(Style <* SELF.Styles |
      'IFC2X3_CV_V2_0.IFCSURFACESTYLEREFRACTION' IN
      TYPEOF(Style)
    )) <= 1;
    WR14 : SIZEOF(QUERY(Style <* SELF.Styles |
      'IFC2X3_CV_V2_0.IFCSURFACESTYLEWITHTEXTURES' IN
      TYPEOF(Style)
    )) <= 1;
    WR15 : SIZEOF(QUERY(Style <* SELF.Styles |
      'IFC2X3_CV_V2_0.IFCEXTERNALLYDEFINEDSURFACESTYLE' IN
      TYPEOF(Style)
    )) <= 1;
END_ENTITY;

```

```

ENTITY IfcSurfaceStyleRendering
  SUBTYPE OF (IfcSurfaceStyleShading);
  Transparency : OPTIONAL IfcNormalisedRatioMeasure;
  DiffuseColour : OPTIONAL IfcColourOrFactor;
  TransmissionColour : OPTIONAL IfcColourOrFactor;
  DiffuseTransmissionColour : OPTIONAL IfcColourOrFactor;
  ReflectionColour : OPTIONAL IfcColourOrFactor;
  SpecularColour : OPTIONAL IfcColourOrFactor;
  SpecularHighlight : OPTIONAL IfcSpecularHighlightSelect;
  ReflectanceMethod : IfcReflectanceMethodEnum;
END_ENTITY;

```

```

ENTITY IfcSurfaceStyleShading

```

```

SUPERTYPE OF (ONEOF
  (IfcSurfaceStyleRendering));
  SurfaceColour : IfcColourRgb;
END_ENTITY;

```

```

ENTITY IfcSweptAreaSolid
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcExtrudedAreaSolid
      ,IfcRevolvedAreaSolid))
  SUBTYPE OF (IfcSolidModel);
    SweptArea : IfcProfileDef;
    Position : IfcAxis2Placement3D;
  WHERE
    WR22 : SweptArea.ProfileType = IfcProfileTypeEnum.Area;
END_ENTITY;

```

```

ENTITY IfcSweptDiskSolid
  SUBTYPE OF (IfcSolidModel);
    Directrix : IfcCurve;
    Radius : IfcPositiveLengthMeasure;
    InnerRadius : OPTIONAL IfcPositiveLengthMeasure;
    StartParam : IfcParameterValue;
    EndParam : IfcParameterValue;
  WHERE
    WR1 : Directrix.Dim = 3;
    WR2 : (NOT EXISTS(InnerRadius)) OR (Radius > InnerRadius);
END_ENTITY;

```

```

ENTITY IfcSwitchingDeviceType
  SUBTYPE OF (IfcFlowControllerType);
    PredefinedType : IfcSwitchingDeviceTypeEnum;
END_ENTITY;

```

```

ENTITY IfcSystem
  SUBTYPE OF (IfcGroup);
  INVERSE
    ServicesBuildings : SET [0:1] OF IfcRelServicesBuildings FOR RelatingSystem;
END_ENTITY;

```

```

ENTITY IfcTShapeProfileDef
  SUBTYPE OF (IfcParameterizedProfileDef);
    Depth : IfcPositiveLengthMeasure;
    FlangeWidth : IfcPositiveLengthMeasure;
    WebThickness : IfcPositiveLengthMeasure;
    FlangeThickness : IfcPositiveLengthMeasure;
    FilletRadius : OPTIONAL IfcPositiveLengthMeasure;
    FlangeEdgeRadius : OPTIONAL IfcPositiveLengthMeasure;
    WebEdgeRadius : OPTIONAL IfcPositiveLengthMeasure;
    WebSlope : OPTIONAL IfcPlaneAngleMeasure;
    FlangeSlope : OPTIONAL IfcPlaneAngleMeasure;
    CentreOfGravityInY : OPTIONAL IfcPositiveLengthMeasure;
  WHERE
    WR1 : FlangeThickness < Depth;
    WR2 : WebThickness < FlangeWidth;
END_ENTITY;

```

```

ENTITY IfcTankType

```

```

SUBTYPE OF (IfcFlowStorageDeviceType);
  PredefinedType : IfcTankTypeEnum;
WHERE
  WR1 : (PredefinedType <> IfcTankTypeEnum.USERDEFINED) OR
        ((PredefinedType = IfcTankTypeEnum.USERDEFINED) AND
         EXISTS(SELf\IfcElementType.ElementType));
END_ENTITY;

```

```

ENTITY IfcTelecomAddress
SUBTYPE OF (IfcAddress);
  TelephoneNumbers : OPTIONAL LIST [1:?] OF IfcLabel;
  FacsimileNumbers : OPTIONAL LIST [1:?] OF IfcLabel;
  PagerNumber : OPTIONAL IfcLabel;
  ElectronicMailAddresses : OPTIONAL LIST [1:?] OF IfcLabel;
  WWWHomePageURL : OPTIONAL IfcLabel;
WHERE
  WR1 : EXISTS (TelephoneNumbers) OR
        EXISTS (PagerNumber) OR
        EXISTS (FacsimileNumbers) OR
        EXISTS (ElectronicMailAddresses) OR
        EXISTS (WWWHomePageURL);
END_ENTITY;

```

```

ENTITY IfcTendon
SUBTYPE OF (IfcReinforcingElement);
  PredefinedType : IfcTendonTypeEnum;
  NominalDiameter : IfcPositiveLengthMeasure;
  CrossSectionArea : IfcAreaMeasure;
  TensionForce : OPTIONAL IfcForceMeasure;
  PreStress : OPTIONAL IfcPressureMeasure;
  FrictionCoefficient : OPTIONAL IfcNormalisedRatioMeasure;
  AnchorageSlip : OPTIONAL IfcPositiveLengthMeasure;
  MinCurvatureRadius : OPTIONAL IfcPositiveLengthMeasure;
WHERE
  WR1 : (PredefinedType <> IfcTendonTypeEnum.USERDEFINED) OR ((PredefinedType =
    IfcTendonTypeEnum.USERDEFINED) AND EXISTS(SELf\IfcObject.ObjectType));
END_ENTITY;

```

```

ENTITY IfcTendonAnchor
SUBTYPE OF (IfcReinforcingElement);
END_ENTITY;

```

```

ENTITY IfcTopologicalRepresentationItem
ABSTRACT SUPERTYPE OF (ONEOF
  (IfcConnectedFaceSet
   ,IfcFace
   ,IfcFaceBound
   ,IfcLoop))
SUBTYPE OF (IfcRepresentationItem);
END_ENTITY;

```

```

ENTITY IfcTransformerType
SUBTYPE OF (IfcEnergyConversionDeviceType);
  PredefinedType : IfcTransformerTypeEnum;
END_ENTITY;

```

```

ENTITY IfcTransportElement

```

```

SUBTYPE OF (IfcElement);
  OperationType : OPTIONAL IfcTransportElementTypeEnum;
  CapacityByWeight : OPTIONAL IfcMassMeasure;
  CapacityByNumber : OPTIONAL IfcCountMeasure;
END_ENTITY;

ENTITY IfcTransportElementType
  SUBTYPE OF (IfcElementType);
    PredefinedType : IfcTransportElementTypeEnum;
END_ENTITY;

ENTITY IfcTrimmedCurve
  SUBTYPE OF (IfcBoundedCurve);
    BasisCurve : IfcCurve;
    Trim1 : SET [1:2] OF IfcTrimmingSelect;
    Trim2 : SET [1:2] OF IfcTrimmingSelect;
    SenseAgreement : BOOLEAN;
    MasterRepresentation : IfcTrimmingPreference;
  WHERE
    WR41 : (HIINDEX(Trim1) = 1) OR (TYPEOF(Trim1[1]) <> TYPEOF(Trim1[2]));
    WR42 : (HIINDEX(Trim2) = 1) OR (TYPEOF(Trim2[1]) <> TYPEOF(Trim2[2]));
    WR43 : NOT('IFC2X3_CV_V2_0.IFCBOUNDED_CURVE' IN TYPEOF(BasisCurve));
END_ENTITY;

ENTITY IfcTubeBundleType
  SUBTYPE OF (IfcEnergyConversionDeviceType);
    PredefinedType : IfcTubeBundleTypeEnum;
  WHERE
    WR1 : (PredefinedType <> IfcTubeBundleTypeEnum.USERDEFINED) OR
      ((PredefinedType = IfcTubeBundleTypeEnum.USERDEFINED) AND
      EXISTS(SELf\IfcElementType.ElementType));
END_ENTITY;

ENTITY IfcTypeObject
  ABSTRACT SUPERTYPE OF (ONEOF
    (IfcTypeProduct))
  SUBTYPE OF (IfcObjectDefinition);
    ApplicableOccurrence : OPTIONAL IfcLabel;
    HasPropertySets : OPTIONAL SET [1:?] OF IfcPropertySetDefinition;
  INVERSE
    ObjectTypeOf : SET [0:1] OF IfcRelDefinesByType FOR RelatingType;
  WHERE
    WR1 : EXISTS(SELf\IfcRoot.Name);
END_ENTITY;

ENTITY IfcTypeProduct
  SUPERTYPE OF (ONEOF
    (IfcDoorStyle
    ,IfcElementType
    ,IfcWindowStyle))
  SUBTYPE OF (IfcTypeObject);
    RepresentationMaps : OPTIONAL LIST [1:?] OF UNIQUE IfcRepresentationMap;
    Tag : OPTIONAL IfcLabel;
  WHERE
    WR41 : NOT(EXISTS(SELf\IfcTypeObject.ObjectTypeOf[1])) OR
      (SIZEOF(QUERY(temp <* SELf\IfcTypeObject.ObjectTypeOf[1].RelatedObjects |
      NOT('IFC2X3_CV_V2_0.IFCPRODUCT' IN TYPEOF(temp))))

```

) = 0);

END_ENTITY;

ENTITY IfcUShapeProfileDef

SUBTYPE OF (IfcParameterizedProfileDef);

Depth : IfcPositiveLengthMeasure;

FlangeWidth : IfcPositiveLengthMeasure;

WebThickness : IfcPositiveLengthMeasure;

FlangeThickness : IfcPositiveLengthMeasure;

FilletRadius : OPTIONAL IfcPositiveLengthMeasure;

EdgeRadius : OPTIONAL IfcPositiveLengthMeasure;

FlangeSlope : OPTIONAL IfcPlaneAngleMeasure;

CentreOfGravityInX : OPTIONAL IfcPositiveLengthMeasure;

WHERE

WR21 : FlangeThickness < (Depth / 2.);

WR22 : WebThickness < FlangeWidth;

END_ENTITY;

ENTITY IfcUnitAssignment;

Units : SET [1:?] OF IfcUnit;

WHERE

WR01 : IfcCorrectUnitAssignment(Units);

END_ENTITY;

ENTITY IfcUnitaryEquipmentType

SUBTYPE OF (IfcEnergyConversionDeviceType);

PredefinedType : IfcUnitaryEquipmentTypeEnum;

WHERE

WR1 : (PredefinedType <> IfcUnitaryEquipmentTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcUnitaryEquipmentTypeEnum.USERDEFINED) AND
 EXISTS(SELf\IfcElementType.ElementType));

END_ENTITY;

ENTITY IfcValveType

SUBTYPE OF (IfcFlowControllerType);

PredefinedType : IfcValveTypeEnum;

WHERE

WR1 : (PredefinedType <> IfcValveTypeEnum.USERDEFINED) OR
 ((PredefinedType = IfcValveTypeEnum.USERDEFINED) AND
 EXISTS(SELf\IfcElementType.ElementType));

END_ENTITY;

ENTITY IfcVector

SUBTYPE OF (IfcGeometricRepresentationItem);

Orientation : IfcDirection;

Magnitude : IfcLengthMeasure;

DERIVE

Dim : IfcDimensionCount := Orientation.Dim;

WHERE

WR1 : Magnitude >= 0.0;

END_ENTITY;

ENTITY IfcWall

SUPERTYPE OF (ONEOF

(IfcWallStandardCase))

SUBTYPE OF (IfcBuildingElement);

WHERE

```

WR1 : SIZEOF (QUERY(temp <* SELF\IfcObjectDefinition.HasAssociations |
    'IFC2X3_CV_V2_0.IFCRELASSOCIATESMATERIAL' IN TYPEOF(temp)
)) <= 1;

```

```

END_ENTITY;

```

```

ENTITY IfcWallStandardCase

```

```

    SUBTYPE OF (IfcWall);

```

```

    WHERE

```

```

        WR1 : SIZEOF (QUERY(temp <* USEDIN(SELF,
            'IFC2X3_CV_V2_0.IFCRELASSOCIATES.RELATEDOBJECTS') |
                ('IFC2X3_CV_V2_0.IFCRELASSOCIATESMATERIAL' IN TYPEOF(temp)) AND
                ('IFC2X3_CV_V2_0.IFCMATERIALLAYERSETUSAGE' IN
                TYPEOF(temp.RelatingMaterial))
            )) = 1;

```

```

END_ENTITY;

```

```

ENTITY IfcWallType

```

```

    SUBTYPE OF (IfcBuildingElementType);

```

```

    PredefinedType : IfcWallTypeEnum;

```

```

END_ENTITY;

```

```

ENTITY IfcWasteTerminalType

```

```

    SUBTYPE OF (IfcFlowTerminalType);

```

```

    PredefinedType : IfcWasteTerminalTypeEnum;

```

```

END_ENTITY;

```

```

ENTITY IfcWindow

```

```

    SUBTYPE OF (IfcBuildingElement);

```

```

    OverallHeight : OPTIONAL IfcPositiveLengthMeasure;

```

```

    OverallWidth : OPTIONAL IfcPositiveLengthMeasure;

```

```

END_ENTITY;

```

```

ENTITY IfcWindowLiningProperties

```

```

    SUBTYPE OF (IfcPropertySetDefinition);

```

```

    LiningDepth : OPTIONAL IfcPositiveLengthMeasure;

```

```

    LiningThickness : OPTIONAL IfcPositiveLengthMeasure;

```

```

    TransomThickness : OPTIONAL IfcPositiveLengthMeasure;

```

```

    MullionThickness : OPTIONAL IfcPositiveLengthMeasure;

```

```

    FirstTransomOffset : OPTIONAL IfcNormalisedRatioMeasure;

```

```

    SecondTransomOffset : OPTIONAL IfcNormalisedRatioMeasure;

```

```

    FirstMullionOffset : OPTIONAL IfcNormalisedRatioMeasure;

```

```

    SecondMullionOffset : OPTIONAL IfcNormalisedRatioMeasure;

```

```

    ShapeAspectStyle : OPTIONAL IfcStrippedOptional;

```

```

    WHERE

```

```

        WR31 : NOT(NOT(EXISTS(LiningDepth)) AND EXISTS(LiningThickness));

```

```

        WR32 : NOT(NOT(EXISTS(FirstTransomOffset)) AND EXISTS(SecondTransomOffset));

```

```

        WR33 : NOT(NOT(EXISTS(FirstMullionOffset)) AND EXISTS(SecondMullionOffset));

```

```

        WR34 : EXISTS(SELF\IfcPropertySetDefinition.DefinesType[1]) AND
            ('IFC2X3_CV_V2_0.IFCWINDOWSTYLE' IN
            TYPEOF(SELF\IfcPropertySetDefinition.DefinesType[1]));

```

```

END_ENTITY;

```

```

ENTITY IfcWindowPanelProperties

```

```

    SUBTYPE OF (IfcPropertySetDefinition);

```

```

    OperationType : IfcWindowPanelOperationEnum;

```

```

    PanelPosition : IfcWindowPanelPositionEnum;

```

```

    FrameDepth : OPTIONAL IfcPositiveLengthMeasure;

```

```

    FrameThickness : OPTIONAL IfcPositiveLengthMeasure;
    ShapeAspectStyle : OPTIONAL IfcStrippedOptional;
END_ENTITY;

```

```

ENTITY IfcWindowStyle
  SUBTYPE OF (IfcTypeProduct);
    ConstructionType : IfcWindowStyleConstructionEnum;
    OperationType : IfcWindowStyleOperationEnum;
    ParameterTakesPrecedence : BOOLEAN;
    Sizeable : BOOLEAN;
END_ENTITY;

```

```

ENTITY IfcZShapeProfileDef
  SUBTYPE OF (IfcParameterizedProfileDef);
    Depth : IfcPositiveLengthMeasure;
    FlangeWidth : IfcPositiveLengthMeasure;
    WebThickness : IfcPositiveLengthMeasure;
    FlangeThickness : IfcPositiveLengthMeasure;
    FilletRadius : OPTIONAL IfcPositiveLengthMeasure;
    EdgeRadius : OPTIONAL IfcPositiveLengthMeasure;
  WHERE
    WR21 : FlangeThickness < (Depth / 2.);
END_ENTITY;

```

```

ENTITY IfcZone
  SUBTYPE OF (IfcGroup);
  WHERE
    WR1 : SIZEOF (QUERY (temp <* SELF\IfcGroup.IsGroupedBy.RelatedObjects |
    NOT(('IFC2X3_CV_V2_0.IFCZONE' IN TYPEOF(temp)) OR ('IFC2X3_CV_V2_0.IFCSPACE' IN
    TYPEOF(temp))) )) = 0;
END_ENTITY;

```

```

FUNCTION IfcBaseAxis
  (Dim : INTEGER;
   Axis1, Axis2, Axis3 : IfcDirection)
  : LIST [2:3] OF IfcDirection;

```

```

  LOCAL
    U : LIST [2:3] OF IfcDirection;
    Factor : REAL;
    D1, D2 : IfcDirection;
  END_LOCAL;

```

```

  IF (Dim = 3) THEN
    D1 := NVL(IfcNormalise(Axis3), IfcRepresentationItem() ||
    IfcGeometricRepresentationItem() || IfcDirection([0.0,0.0,1.0]));
    D2 := IfcFirstProjAxis(D1, Axis1);
    U := [D2, IfcSecondProjAxis(D1, D2, Axis2), D1];
  ELSE
    IF EXISTS(Axis1) THEN
      D1 := IfcNormalise(Axis1);
      U := [D1, IfcOrthogonalComplement(D1)];
      IF EXISTS(Axis2) THEN
        Factor := IfcDotProduct(Axis2, U[2]);
        IF (Factor < 0.0) THEN
          U[2].DirectionRatios[1] := -U[2].DirectionRatios[1];
          U[2].DirectionRatios[2] := -U[2].DirectionRatios[2];

```

```

        END_IF;
    END_IF;
ELSE
    IF EXISTS(Axis2) THEN
        D1 := IfcNormalise(Axis2);
        U := [IfcOrthogonalComplement(D1), D1];
        U[1].DirectionRatios[1] := -U[1].DirectionRatios[1];
        U[1].DirectionRatios[2] := -U[1].DirectionRatios[2];
    ELSE
        U := [IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
            IfcDirection([1.0, 0.0]),
            IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
            IfcDirection([0.0, 1.0])];
    END_IF;
END_IF;
END_IF;
RETURN(U);
END_FUNCTION;

FUNCTION IfcBuild2Axes
    (RefDirection : IfcDirection)
    : LIST [2:2] OF IfcDirection;
LOCAL
    D : IfcDirection := NVL(IfcNormalise(RefDirection),
        IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcDirection([1.0,0.0]));
END_LOCAL;
RETURN([D, IfcOrthogonalComplement(D)]);
END_FUNCTION;

FUNCTION IfcBuildAxes
    (Axis, RefDirection : IfcDirection)
    : LIST [3:3] OF IfcDirection;
LOCAL
    D1, D2 : IfcDirection;
END_LOCAL;
D1 := NVL(IfcNormalise(Axis), IfcRepresentationItem() || IfcGeometricRepresentationItem
    () || IfcDirection([0.0,0.0,1.0]));
D2 := IfcFirstProjAxis(D1, RefDirection);
RETURN ([D2, IfcNormalise(IfcCrossProduct(D1,D2))\IfcVector.Orientation, D1]);
END_FUNCTION;

FUNCTION IfcCorrectDimensions
    (m : IfcUnitEnum; Dim : IfcDimensionalExponents)
    : LOGICAL;
CASE m OF
    LENGTHUNIT : IF
        Dim = (IfcDimensionalExponents (1, 0, 0, 0, 0, 0, 0))
        THEN RETURN(TRUE);
        ELSE RETURN(FALSE);
    END_IF;
    MASSUNIT : IF
        Dim = (IfcDimensionalExponents (0, 1, 0, 0, 0, 0, 0))
        THEN RETURN(TRUE);
        ELSE RETURN(FALSE);
    END_IF;
    TIMEUNIT : IF

```

```

Dim = (IfcDimensionalExponents (0, 0, 1, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
ELECTRICCURRENTUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 1, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
THERMODYNAMICTEMPERATUREUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 0, 1, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
AMOUNTOFSUBSTANCEUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 0, 0, 1, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
LUMINOUSINTENSITYUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 0, 0, 0, 1))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
PLANEANGLEUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
SOLIDANGLEUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, 0, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
AREAUNIT : IF
Dim = (IfcDimensionalExponents (2, 0, 0, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
VOLUMEUNIT : IF
Dim = (IfcDimensionalExponents (3, 0, 0, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;

ABSORBEDDOSEUNIT : IF
Dim = (IfcDimensionalExponents (2, 0, -2, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
RADIOACTIVITYUNIT : IF
Dim = (IfcDimensionalExponents (0, 0, -1, 0, 0, 0, 0))
THEN RETURN(TRUE);
ELSE RETURN(FALSE);
END_IF;
ELECTRICCAPACITANCEUNIT : IF
Dim = (IfcDimensionalExponents (-2, 1, 4, 1, 0, 0, 0))

```

```

    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
DOSEEQUIVALENTUNIT : IF
    Dim = (IfcDimensionalExponents (2, 0, -2, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ELECTRICCHARGEUNIT : IF
    Dim = (IfcDimensionalExponents (0, 0, 1, 1, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ELECTRICCONDUCTANCEUNIT : IF
    Dim = (IfcDimensionalExponents (-2, -1, 3, 2, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ELECTRICVOLTAGEUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -3, -1, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ELECTRICRESISTANCEUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -3, -2, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ENERGYUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -2, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
FORCEUNIT : IF
    Dim = (IfcDimensionalExponents (1, 1, -2, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
FREQUENCYUNIT : IF
    Dim = (IfcDimensionalExponents (0, 0, -1, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
INDUCTANCEUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -2, -2, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
ILLUMINANCEUNIT : IF
    Dim = (IfcDimensionalExponents (-2, 0, 0, 0, 0, 0, 1))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
END_IF;
LUMINOUSFLUXUNIT : IF
    Dim = (IfcDimensionalExponents (0, 0, 0, 0, 0, 0, 1))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);

```

```

    END_IF;
MAGNETICFLUXUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -2, -1, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
    END_IF;
MAGNETICFLUXDENSITYUNIT : IF
    Dim = (IfcDimensionalExponents (0, 1, -2, -1, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
    END_IF;
POWERUNIT : IF
    Dim = (IfcDimensionalExponents (2, 1, -3, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
    END_IF;
PRESSUREUNIT : IF
    Dim = (IfcDimensionalExponents (-1, 1, -2, 0, 0, 0, 0))
    THEN RETURN(TRUE);
    ELSE RETURN(FALSE);
    END_IF;

    OTHERWISE :
        RETURN (UNKNOWN);
END_CASE;
END_FUNCTION;

FUNCTION IfcCorrectLocalPlacement
    (AxisPlacement:IfcAxis2Placement;
    RelPlacement : IfcObjectPlacement)
    : LOGICAL;

    IF (EXISTS(RelPlacement)) THEN
        IF ('IFC2X3_CV_V2_0.IFCGRIDPLACEMENT' IN TYPEOF(RelPlacement)) THEN
            RETURN(?);
        END_IF;
        IF ('IFC2X3_CV_V2_0.IFCLOCALPLACEMENT' IN TYPEOF(RelPlacement)) THEN
            IF ('IFC2X3_CV_V2_0.IFCAXIS2PLACEMENT2D' IN TYPEOF(AxisPlacement)) THEN
                RETURN(TRUE);
            END_IF;
            IF ('IFC2X3_CV_V2_0.IFCAXIS2PLACEMENT3D' IN TYPEOF(AxisPlacement)) THEN
                IF (RelPlacement.RelativePlacement.Dim = 3) THEN
                    RETURN(TRUE);
                ELSE
                    RETURN(FALSE);
                END_IF;
            END_IF;
        END_IF;
    ELSE
        RETURN(TRUE);
    END_IF;
    RETURN(?);
END_FUNCTION;

FUNCTION IfcCorrectObjectAssignment
    (Constraint: IfcObjectTypeEnum; Objects : SET[1:?] OF IfcObjectDefinition)
    : LOGICAL;

```

```

LOCAL
  Count : INTEGER := 0;
END_LOCAL;

IF NOT(EXISTS(Constraint)) THEN
  RETURN(TRUE);
END_IF;

CASE Constraint OF
  IfcObjectTypeEnum.NOTDEFINED : RETURN(TRUE);
  IfcObjectTypeEnum.PRODUCT :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCPRODUCT' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.PROCESS :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCPROCESS' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.CONTROL :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCCONTROL' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.RESOURCE :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCRESOURCE' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.ACTOR :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCACTOR' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.GROUP :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCGROUP' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  IfcObjectTypeEnum.PROJECT :
    BEGIN
      Count := SIZEOF(QUERY(temp <* Objects | NOT('IFC2X3_CV_V2_0.IFCPROJECT' IN
        TYPEOF(temp))));
      RETURN(Count = 0);
    END;
  OTHERWISE : RETURN(?);
END_CASE;
END_FUNCTION;

```

```

FUNCTION IfcCorrectUnitAssignment
  (Units : SET [1:?] OF IfcUnit)
  : LOGICAL;

LOCAL
  NamedUnitNumber      : INTEGER := 0;
  DerivedUnitNumber    : INTEGER := 0;
  MonetaryUnitNumber   : INTEGER := 0;
  NamedUnitNames       : SET OF IfcUnitEnum := [];
  DerivedUnitNames     : SET OF IfcDerivedUnitEnum := [];
END_LOCAL;

NamedUnitNumber      := SIZEOF(QUERY(temp <* Units | ('IFC2X3_CV_V2_0.IFCNAMEDUNIT' IN
  TYPEOF(temp)) AND NOT(temp\IfcNamedUnit.UnitType = IfcUnitEnum.USERDEFINED)));
DerivedUnitNumber    := SIZEOF(QUERY(temp <* Units | ('IFC2X3_CV_V2_0.IFCDERIVEDUNIT' IN
  TYPEOF(temp)) AND NOT(temp\IfcDerivedUnit.UnitType = IfcDerivedUnitEnum.USERDEFINED)));
MonetaryUnitNumber   := SIZEOF(QUERY(temp <* Units | 'IFC2X3_CV_V2_0.IFCMONETARYUNIT' IN
  TYPEOF(temp)));

REPEAT i := 1 TO SIZEOF(Units);
  IF (('IFC2X3_CV_V2_0.IFCNAMEDUNIT' IN TYPEOF(Units[i])) AND
    NOT(Units[i]\IfcNamedUnit.UnitType = IfcUnitEnum.USERDEFINED)) THEN
    NamedUnitNames := NamedUnitNames + Units[i]\IfcNamedUnit.UnitType;
  END_IF;
  IF (('IFC2X3_CV_V2_0.IFCDERIVEDUNIT' IN TYPEOF(Units[i])) AND
    NOT(Units[i]\IfcDerivedUnit.UnitType = IfcDerivedUnitEnum.USERDEFINED)) THEN
    DerivedUnitNames := DerivedUnitNames + Units[i]\IfcDerivedUnit.UnitType;
  END_IF;
END_REPEAT;

RETURN((SIZEOF(NamedUnitNames) = NamedUnitNumber) AND (SIZEOF(DerivedUnitNames) =
  DerivedUnitNumber) AND (MonetaryUnitNumber <= 1));
END_FUNCTION;

FUNCTION IfcCrossProduct
  (Arg1, Arg2 : IfcDirection)
  : IfcVector;

LOCAL
  Mag : REAL;
  Res : IfcDirection;
  V1,V2 : LIST[3:3] OF REAL;
  Result : IfcVector;
END_LOCAL;

IF (NOT EXISTS (Arg1) OR (Arg1.Dim = 2)) OR (NOT EXISTS (Arg2) OR (Arg2.Dim = 2)) THEN
  RETURN(?);
ELSE
  BEGIN
    V1 := IfcNormalise(Arg1).DirectionRatios;
    V2 := IfcNormalise(Arg2).DirectionRatios;
    Res := IfcRepresentationItem() || IfcGeometricRepresentationItem (
      || IfcDirection([(V1[2]*V2[3] - V1[3]*V2[2]), (V1[3]*V2[1] - V1[1]*V2[3]),
        (V1[1]*V2[2] - V1[2]*V2[1])]);
    Mag := 0.0;
    REPEAT i := 1 TO 3;
      Mag := Mag + Res.DirectionRatios[i]*Res.DirectionRatios[i];
    END_REPEAT;
  END
END

```

```

    IF (Mag > 0.0) THEN
        Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcVector(Res, SQRT(Mag));
    ELSE
        Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcVector(Arg1, 0.0);
    END_IF;
    RETURN(Result);
END;
END_IF;
END_FUNCTION;

```

```

FUNCTION IfcCurveDim
    (Curve : IfcCurve)
    : IfcDimensionCount;

    IF ('IFC2X3_CV_V2_0.IFCLINE' IN TYPEOF(Curve))
        THEN RETURN(Curve\IfcLine.Pnt.Dim);
    END_IF;
    IF ('IFC2X3_CV_V2_0.IFCCONIC' IN TYPEOF(Curve))
        THEN RETURN(Curve\IfcConic.Position.Dim);
    END_IF;
    IF ('IFC2X3_CV_V2_0.IFCPOLYLINE' IN TYPEOF(Curve))
        THEN RETURN(Curve\IfcPolyline.Points[1].Dim);
    END_IF;
    IF ('IFC2X3_CV_V2_0.IFCTRIMMEDCURVE' IN TYPEOF(Curve))
        THEN RETURN(IfcCurveDim(Curve\IfcTrimmedCurve.BasisCurve));
    END_IF;
    IF ('IFC2X3_CV_V2_0.IFCCOMPOSITECURVE' IN TYPEOF(Curve))
        THEN RETURN(Curve\IfcCompositeCurve.Segments[1].Dim);
    END_IF;
    RETURN (?);
END_FUNCTION;

```

```

FUNCTION IfcDeriveDimensionalExponents
    (UnitElements : SET [1:?] OF IfcDerivedUnitElement)
    : IfcDimensionalExponents;
    LOCAL
        Result : IfcDimensionalExponents :=
            IfcDimensionalExponents(0, 0, 0, 0, 0, 0, 0);
    END_LOCAL;
    REPEAT i := LOINDEX(UnitElements) TO HIINDEX(UnitElements);
        Result.LengthExponent := Result.LengthExponent +
            (UnitElements[i].Exponent *
            UnitElements[i].Unit.Dimensions.LengthExponent);
        Result.MassExponent := Result.MassExponent +
            (UnitElements[i].Exponent *
            UnitElements[i].Unit.Dimensions.MassExponent);
        Result.TimeExponent := Result.TimeExponent +
            (UnitElements[i].Exponent *
            UnitElements[i].Unit.Dimensions.TimeExponent);
        Result.ElectricCurrentExponent := Result.ElectricCurrentExponent +
            (UnitElements[i].Exponent *
            UnitElements[i].Unit.Dimensions.ElectricCurrentExponent);
        Result.ThermodynamicTemperatureExponent := Result.ThermodynamicTemperatureExponent +
            (UnitElements[i].Exponent *

```

```

        UnitElements[i].Unit.Dimensions.ThermodynamicTemperatureExponent);
    Result.AmountOfSubstanceExponent := Result.AmountOfSubstanceExponent +
        (UnitElements[i].Exponent *
        UnitElements[i].Unit.Dimensions.AmountOfSubstanceExponent);
    Result.LuminousIntensityExponent := Result.LuminousIntensityExponent +
        (UnitElements[i].Exponent *
        UnitElements[i].Unit.Dimensions.LuminousIntensityExponent);
END_REPEAT;
RETURN (Result);
END_FUNCTION;

```

FUNCTION IfcDimensionsForSiUnit

```

(n : IfcSiUnitName )
: IfcDimensionalExponents;
CASE n OF
    METRE          : RETURN (IfcDimensionalExponents
                            (1, 0, 0, 0, 0, 0, 0));
    SQUARE_METRE   : RETURN (IfcDimensionalExponents
                            (2, 0, 0, 0, 0, 0, 0));
    CUBIC_METRE    : RETURN (IfcDimensionalExponents
                            (3, 0, 0, 0, 0, 0, 0));
    GRAM           : RETURN (IfcDimensionalExponents
                            (0, 1, 0, 0, 0, 0, 0));
    SECOND         : RETURN (IfcDimensionalExponents
                            (0, 0, 1, 0, 0, 0, 0));
    AMPERE         : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 1, 0, 0, 0));
    KELVIN         : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 0, 1, 0, 0));
    MOLE           : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 0, 0, 1, 0));
    CANDELA        : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 0, 0, 0, 1));
    RADIAN         : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 0, 0, 0, 0));
    STERADIAN      : RETURN (IfcDimensionalExponents
                            (0, 0, 0, 0, 0, 0, 0));
    HERTZ          : RETURN (IfcDimensionalExponents
                            (0, 0, -1, 0, 0, 0, 0));
    NEWTON         : RETURN (IfcDimensionalExponents
                            (1, 1, -2, 0, 0, 0, 0));
    PASCAL         : RETURN (IfcDimensionalExponents
                            (-1, 1, -2, 0, 0, 0, 0));
    JOULE          : RETURN (IfcDimensionalExponents
                            (2, 1, -2, 0, 0, 0, 0));
    WATT           : RETURN (IfcDimensionalExponents
                            (2, 1, -3, 0, 0, 0, 0));
    COULOMB        : RETURN (IfcDimensionalExponents
                            (0, 0, 1, 1, 0, 0, 0));
    VOLT           : RETURN (IfcDimensionalExponents
                            (2, 1, -3, -1, 0, 0, 0));
    FARAD          : RETURN (IfcDimensionalExponents
                            (-2, -1, 4, 1, 0, 0, 0));
    OHM            : RETURN (IfcDimensionalExponents
                            (2, 1, -3, -2, 0, 0, 0));
    SIEMENS        : RETURN (IfcDimensionalExponents
                            (-2, -1, 3, 2, 0, 0, 0));

```

```

    WEBER          : RETURN (IfcDimensionalExponents
                           (2, 1, -2, -1, 0, 0, 0));
    TESLA          : RETURN (IfcDimensionalExponents
                           (0, 1, -2, -1, 0, 0, 0));
    HENRY          : RETURN (IfcDimensionalExponents
                           (2, 1, -2, -2, 0, 0, 0));
    DEGREE_CELSIUS : RETURN (IfcDimensionalExponents
                           (0, 0, 0, 0, 1, 0, 0));
    LUMEN          : RETURN (IfcDimensionalExponents
                           (0, 0, 0, 0, 0, 0, 1));
    LUX            : RETURN (IfcDimensionalExponents
                           (-2, 0, 0, 0, 0, 0, 1));
    BECQUEREL      : RETURN (IfcDimensionalExponents
                           (0, 0, -1, 0, 0, 0, 0));
    GRAY           : RETURN (IfcDimensionalExponents
                           (2, 0, -2, 0, 0, 0, 0));
    SIEVERT        : RETURN (IfcDimensionalExponents
                           (2, 0, -2, 0, 0, 0, 0));
    OTHERWISE      : RETURN (IfcDimensionalExponents
                           (0, 0, 0, 0, 0, 0, 0));

    END_CASE;
END_FUNCTION;

FUNCTION IfcDotProduct
  (Arg1, Arg2 : IfcDirection)
  : REAL;
LOCAL
  Scalar : REAL;
  Vec1, Vec2 : IfcDirection;
  Ndim : INTEGER;
END_LOCAL;

IF NOT EXISTS (Arg1) OR NOT EXISTS (Arg2) THEN
  Scalar := ?;
ELSE
  IF (Arg1.Dim <> Arg2.Dim) THEN
    Scalar := ?;
  ELSE
    BEGIN
      Vec1 := IfcNormalise(Arg1);
      Vec2 := IfcNormalise(Arg2);
      Ndim := Arg1.Dim;
      Scalar := 0.0;
      REPEAT i := 1 TO Ndim;
        Scalar := Scalar + Vec1.DirectionRatios[i]*Vec2.DirectionRatios[i];
      END_REPEAT;
    END;
  END_IF;
END_IF;
RETURN (Scalar);
END_FUNCTION;

FUNCTION IfcFirstProjAxis
  (ZAxis, Arg : IfcDirection)
  : IfcDirection;
LOCAL
  XAxis : IfcDirection;

```

```

V      : IfcDirection;
Z      : IfcDirection;
XVec   : IfcVector;
END_LOCAL;

IF (NOT EXISTS(ZAxis)) THEN
  RETURN (?) ;
ELSE
  Z := IfcNormalise(ZAxis);
  IF NOT EXISTS(Arg) THEN
    IF (Z.DirectionRatios <> [1.0,0.0,0.0]) THEN
      V := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcDirection([1.0,0.0,0.0]);
    ELSE
      V := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
        IfcDirection([0.0,1.0,0.0]);
    END_IF;
  ELSE
    IF (Arg.Dim <> 3) THEN
      RETURN (?) ;
    END_IF;
    IF ((IfcCrossProduct(Arg,Z).Magnitude) = 0.0) THEN
      RETURN (?);
    ELSE
      V := IfcNormalise(Arg);
    END_IF;
  END_IF;
  XVec := IfcScalarTimesVector(IfcDotProduct(V, Z), Z);
  XAxis := IfcVectorDifference(V, XVec).Orientation;
  XAxis := IfcNormalise(XAxis);
END_IF;
RETURN(XAxis);
END_FUNCTION;

FUNCTION IfcMlsTotalThickness
  (LayerSet : IfcMaterialLayerSet)
  : IfcLengthMeasure;
LOCAL
  Max : IfcLengthMeasure := LayerSet.MaterialLayers[1].LayerThickness;
END_LOCAL;

IF SIZEOF(LayerSet.MaterialLayers) > 1 THEN
  REPEAT i := 2 TO HIINDEX(LayerSet.MaterialLayers);
    Max := Max + LayerSet.MaterialLayers[i].LayerThickness;
  END_REPEAT;
END_IF;
RETURN (Max);
END_FUNCTION;

FUNCTION IfcNormalise
  (Arg : IfcVectorOrDirection)
  : IfcVectorOrDirection;
LOCAL
  Ndim : INTEGER;
  V      : IfcDirection
    := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
      IfcDirection([1.,0.]);

```

```

Vec  : IfcVector
      := IfcRepresentationItem() || IfcGeometricRepresentationItem () || IfcVector (
          IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
          IfcDirection([1.,0.]), 1.);
Mag  : REAL;
Result : IfcVectorOrDirection
      := V;
END_LOCAL;

IF NOT EXISTS (Arg) THEN
  RETURN (?);
ELSE
  Ndim := Arg.Dim;
  IF 'IFC2X3_CV_V2_0.IFCVECTOR' IN TYPEOF(Arg) THEN
    BEGIN
V.DirectionRatios := Arg\IfcVector.Orientation.DirectionRatios;
    Vec.Magnitude := Arg\IfcVector.Magnitude;
Vec.Orientation := V;
    IF Arg.Magnitude = 0.0 THEN
      RETURN(?);
    ELSE
      Vec.Magnitude := 1.0;
    END_IF;
  END;
ELSE
  V.DirectionRatios := Arg.DirectionRatios;
END_IF;
Mag := 0.0;
  REPEAT i := 1 TO Ndim;
    Mag := Mag + V.DirectionRatios[i]*V.DirectionRatios[i];
  END_REPEAT;
IF Mag > 0.0 THEN
  Mag := SQRT(Mag);
  REPEAT i := 1 TO Ndim;
    V.DirectionRatios[i] := V.DirectionRatios[i]/Mag;
  END_REPEAT;
IF 'IFC2X3_CV_V2_0.IFCVECTOR' IN TYPEOF(arg) THEN
  Vec.Orientation := V;
  Result := Vec;
ELSE
  Result := V;
END_IF;
ELSE
  RETURN(?);
END_IF;
END_IF;
RETURN (Result);
END_FUNCTION;

FUNCTION IfcOrthogonalComplement
  (Vec : IfcDirection)
  : IfcDirection;
LOCAL
  Result : IfcDirection ;
END_LOCAL;
IF NOT EXISTS (Vec) OR (Vec.Dim <> 2) THEN
  RETURN(?);

```

```

ELSE
    Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
    IfcDirection([-Vec.DirectionRatios[2], Vec.DirectionRatios[1]]);
RETURN(Result);
END_IF;
END_FUNCTION;

```

```

FUNCTION IfcScalarTimesVector
(Scalar : REAL; Vec : IfcVectorOrDirection)
: IfcVector;
LOCAL
    V : IfcDirection;
    Mag : REAL;
    Result : IfcVector;
END_LOCAL;

IF NOT EXISTS (Scalar) OR NOT EXISTS (Vec) THEN
    RETURN (?) ;
ELSE
    IF 'IFC2X3_CV_V2_0.IFCVECTOR' IN TYPEOF (Vec) THEN
        V := Vec\IfcVector.Orientation;
        Mag := Scalar * Vec\IfcVector.Magnitude;
    ELSE
        V := Vec;
        Mag := Scalar;
    END_IF;
    IF (Mag < 0.0 ) THEN
        REPEAT i := 1 TO SIZEOF(V.DirectionRatios);
            V.DirectionRatios[i] := -V.DirectionRatios[i];
        END_REPEAT;
        Mag := -Mag;
    END_IF;
    Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
    IfcVector(IfcNormalise(V), Mag);
END_IF;
RETURN (Result);
END_FUNCTION;

```

```

FUNCTION IfcSecondProjAxis
(ZAxis, XAxis, Arg: IfcDirection)
: IfcDirection;
LOCAL
    YAxis : IfcVector;
    V      : IfcDirection;
    Temp   : IfcVector;
END_LOCAL;

IF NOT EXISTS(Arg) THEN
    V := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
    IfcDirection([0.0,1.0,0.0]);
ELSE
    V := Arg;
END_IF;
Temp := IfcScalarTimesVector(IfcDotProduct(V, ZAxis), ZAxis);
YAxis := IfcVectorDifference(V, Temp);
Temp := IfcScalarTimesVector(IfcDotProduct(V, XAxis), XAxis);
YAxis := IfcVectorDifference(YAxis, Temp);

```

```

YAxis := IfcNormalise(YAxis);
RETURN(YAxis.Orientation);
END_FUNCTION;

FUNCTION IfcShapeRepresentationTypes
(RepType : IfcLabel; Items : SET OF IfcRepresentationItem)
: LOGICAL;

LOCAL
Count : INTEGER := 0;
END_LOCAL;

CASE RepType OF
'Curve2D' :
BEGIN
Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCCURVE' IN TYPEOF(temp))
AND (temp.Dim = 2)));
END;
'Annotation2D' :
BEGIN
Count := SIZEOF(QUERY(temp <* Items | (
SIZEOF(TYPEOF(temp) * [
'IFC2X3_CV_V2_0.IFCPOINT',
'IFC2X3_CV_V2_0.IFCCURVE',
'IFC2X3_CV_V2_0.IFCGEOMETRICCURVESET',
'IFC2X3_CV_V2_0.IFCANNOTATIONFILLAREA',
'IFC2X3_CV_V2_0.IFCDEFINEDSYMBOL',
'IFC2X3_CV_V2_0.IFCTEXTLITERAL',
'IFC2X3_CV_V2_0.IFCDRAUGHTINGCALLOUT']) = 1)
)));
END;
'GeometricSet' :
BEGIN
Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCGEOMETRICSET' IN
TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCPOINT' IN TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCCURVE' IN TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCSURFACE' IN TYPEOF(temp))));
END;
'GeometricCurveSet' :
BEGIN
Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCGEOMETRICCURVESET' IN
TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCGEOMETRICSET' IN TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCPOINT' IN TYPEOF(temp))
OR ('IFC2X3_CV_V2_0.IFCCURVE' IN TYPEOF(temp))));
REPEAT i:=1 TO HIINDEX(Items);
IF ('IFC2X3_CV_V2_0.IFCGEOMETRICSET' IN TYPEOF(Items[i]))
THEN
IF (SIZEOF(QUERY(temp <* Items[i]\IfcGeometricSet.Elements |
'IFC2X3_CV_V2_0.IFCSURFACE' IN TYPEOF(temp))) > 0)
THEN
Count := Count - 1;
END_IF;
END_IF;
END_REPEAT;
END;

```

```

'SurfaceModel' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | SIZEOF([
    'IFC2X3_CV_V2_0.IFCSHELLBASEDSURFACEMODEL',
    'IFC2X3_CV_V2_0.IFCFACEBASEDSURFACEMODEL',
    'IFC2X3_CV_V2_0.IFCFACETEDBREP',
    'IFC2X3_CV_V2_0.IFCFACETEDBREPWITHVOIDS'] * TYPEOF(temp)) >= 1
  ));
END;
'SolidModel' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCSOLIDMODEL' IN
    TYPEOF(temp))
    OR ('IFC2X3_CV_V2_0.IFCBOOLEANRESULT' IN TYPEOF(temp))));
END;
'SweptSolid' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCSWEPTAREASOLID' IN
    TYPEOF(temp))));
END;
'CSG' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCBOOLEANRESULT' IN
    TYPEOF(temp))));
END;
'Clipping' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCBOOLEANCLIPPINGRESULT'
    IN TYPEOF(temp))));
END;
'AdvancedSweptSolid' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCSWEPTDISKSOLID' IN
    TYPEOF(temp))));
END;
'Brep' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCFACETEDBREP' IN
    TYPEOF(temp))
    OR ('IFC2X3_CV_V2_0.IFCFACETEDBREPWITHVOIDS' IN TYPEOF(temp))));
END;
'BoundingBox' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCBOUNDINGBOX' IN
    TYPEOF(temp))));
  IF (SIZEOF(Items) > 1)
  THEN
    Count := 0;
  END_IF;
END;
'SectionedSpine' :
BEGIN
  Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCSECTIONEDSPINE' IN
    TYPEOF(temp))));
END;
'MappedRepresentation' :
BEGIN

```

```

        Count := SIZEOF(QUERY(temp <* Items | ('IFC2X3_CV_V2_0.IFCMAPPEDITEM' IN
        TYPEOF(temp))));
    END;
    OTHERWISE : RETURN(?);
END_CASE;
RETURN (Count = SIZEOF(Items));
END_FUNCTION;

```

```

FUNCTION IfcUniquePropertyName
  (Properties : SET [1:?] OF IfcProperty)
  : LOGICAL;

  LOCAL
    Names : SET OF IfcIdentifier := [];
  END_LOCAL;

  REPEAT i:=1 TO HIINDEX(Properties);
    Names := Names + Properties[i].Name;
  END_REPEAT;
  RETURN (SIZEOF(Names) = SIZEOF(Properties));
END_FUNCTION;

```

```

FUNCTION IfcVectorDifference
  (Arg1, Arg2 : IfcVectorOrDirection)
  : IfcVector;
  LOCAL
    Result : IfcVector;
    Res, Vec1, Vec2 : IfcDirection;
    Mag, Mag1, Mag2 : REAL;
    Ndim : INTEGER;
  END_LOCAL;

  IF ((NOT EXISTS (Arg1)) OR (NOT EXISTS (Arg2))) OR (Arg1.Dim <> Arg2.Dim) THEN
    RETURN (?);
  ELSE
    BEGIN
      IF 'IFC2X3_CV_V2_0.IFCVECTOR' IN TYPEOF(Arg1) THEN
        Mag1 := Arg1\IfcVector.Magnitude;
        Vec1 := Arg1\IfcVector.Orientation;
      ELSE
        Mag1 := 1.0;
        Vec1 := Arg1;
      END_IF;
      IF 'IFC2X3_CV_V2_0.IFCVECTOR' IN TYPEOF(Arg2) THEN
        Mag2 := Arg2\IfcVector.Magnitude;
        Vec2 := Arg2\IfcVector.Orientation;
      ELSE
        Mag2 := 1.0;
        Vec2 := Arg2;
      END_IF;
      Vec1 := IfcNormalise (Vec1);
      Vec2 := IfcNormalise (Vec2);
      Ndim := SIZEOF(Vec1.DirectionRatios);
      Mag := 0.0;
      Res := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
      IfcDirection([0.0:Ndim]);
    END
  END

```

```
REPEAT i := 1 TO Ndim;
  Res.DirectionRatios[i] := Mag1*Vec1.DirectionRatios[i] -
  Mag2*Vec2.DirectionRatios[i];
  Mag := Mag + (Res.DirectionRatios[i]*Res.DirectionRatios[i]);
END_REPEAT;

IF (Mag > 0.0 ) THEN
  Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
  IfcVector( Res, SQRT(Mag));
ELSE
  Result := IfcRepresentationItem() || IfcGeometricRepresentationItem () ||
  IfcVector( Vec1, 0.0);
END_IF;
END;
END_IF;
RETURN (Result);
END_FUNCTION;

END_SCHEMA;
```